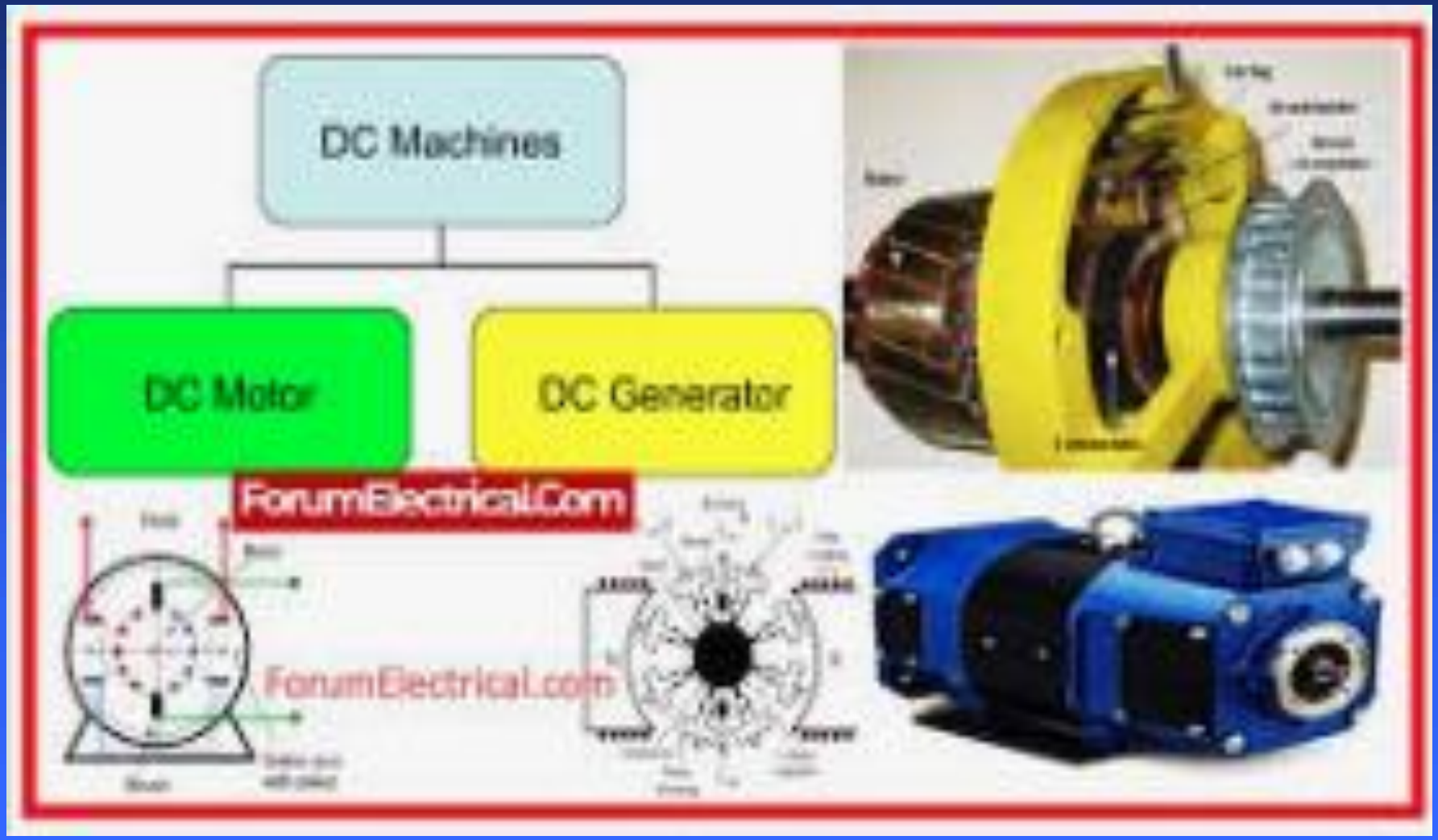
Lecture 2: DC Machinery Fundamentals



Preliminary notes

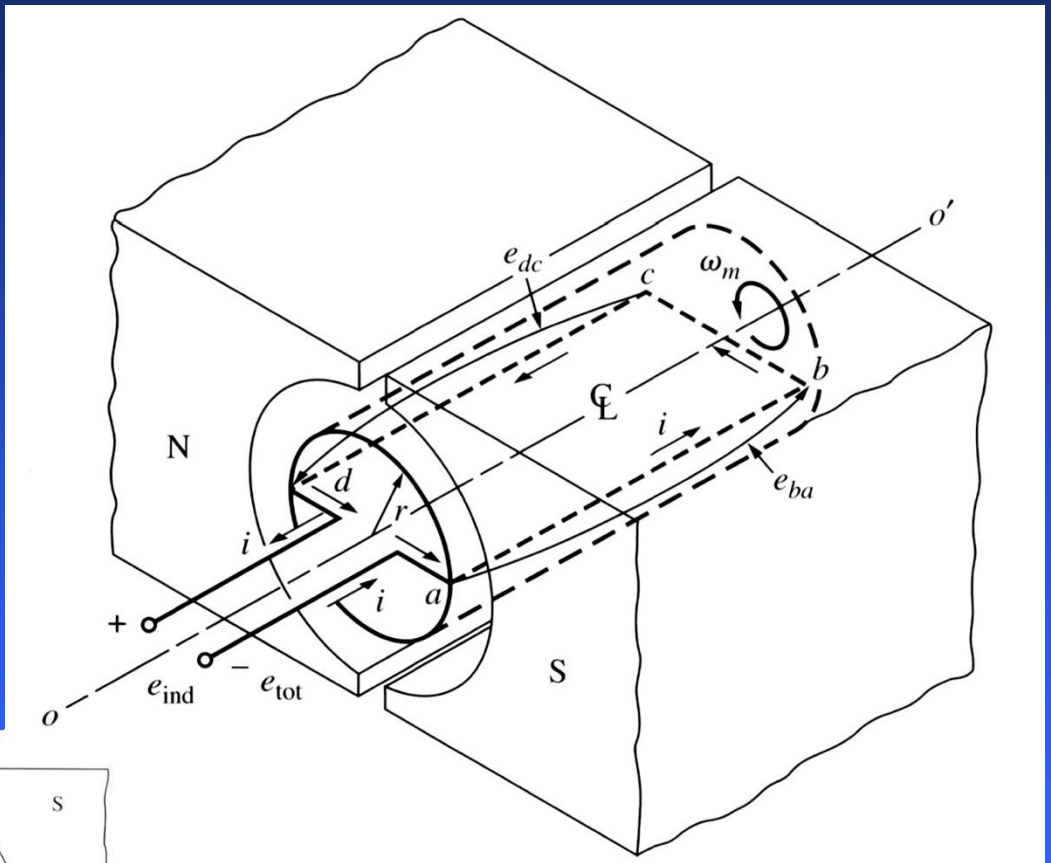
- DC power systems are not very common in the contemporary engineering practice. However, DC motors still have many practical applications, such automobile, aircraft, and portable electronics, in speed control applications…
- An advantage of DC motors is that it is easy to control their speed in a wide range.
- DC generators are quite rare.
- Most DC machines are similar to AC machines: i.e. they have AC voltages and current within them. DC machines have DC outputs just because they have a mechanism converting AC voltages to DC voltages at their terminals. This mechanism is called a commutator; therefore, DC machines are also called commutating machines.

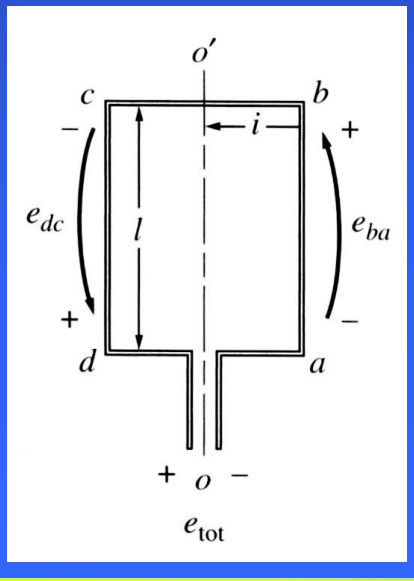


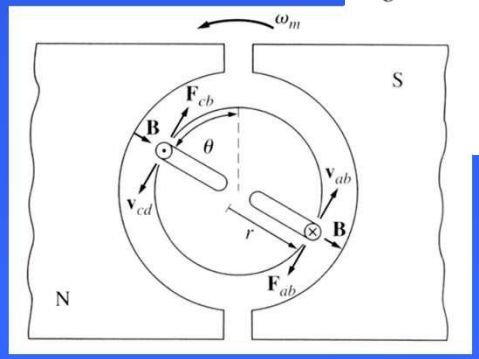


The simplest DC machine

The simplest DC rotating machine consists of a single loop of wire rotating about a fixed axis. The magnetic field is supplied by the North and South poles of the magnet.
Rotor is the rotating part;
Stator is the stationary part.



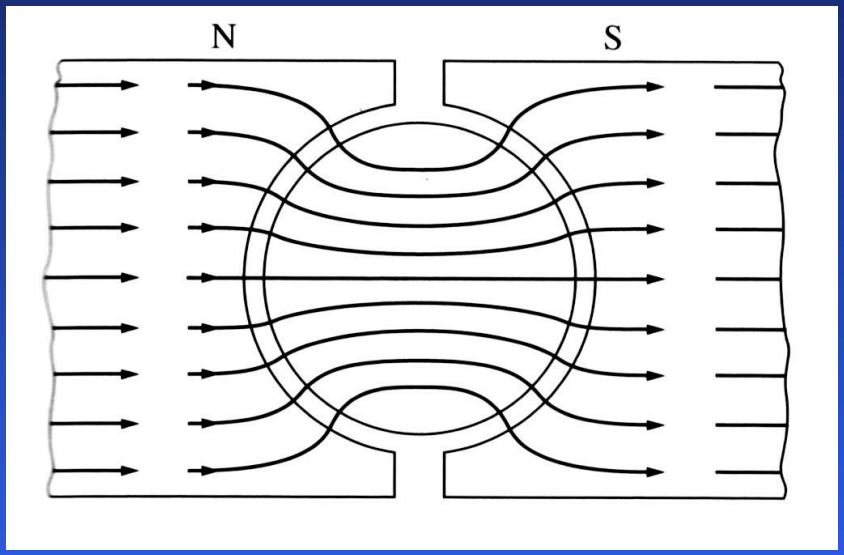




The simplest DC machine

We notice that the rotor lies in a slot curved in a ferromagnetic stator core, which, together with the rotor core, provides a constant-width air gap between the rotor and stator.

The reluctance of air is much larger than the reluctance of core. Therefore, the magnetic flux must take the shortest path through the air gap.



As a consequence, the magnetic flux is perpendicular to the rotor surface everywhere under the pole faces.

Since the air gap is uniform, the reluctance is constant everywhere under the pole faces. Therefore, magnetic flux density is also constant everywhere under the pole faces.

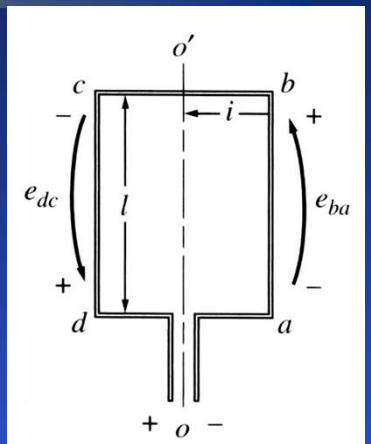
The simplest DC machine

1. Voltage induced in a rotating loop

If a rotor of a DC machine is rotated, a voltage will be induced…

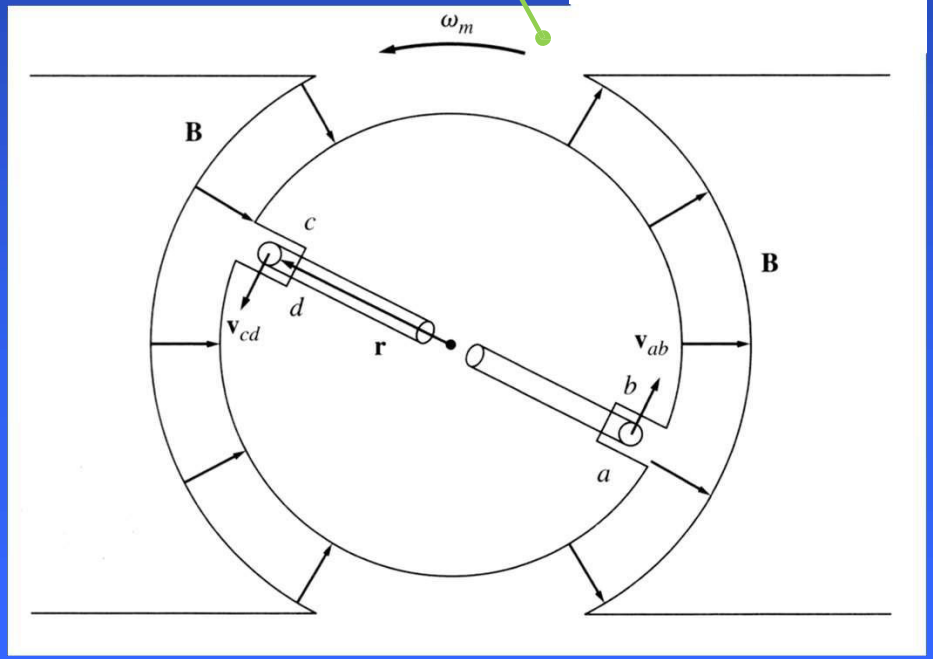
The loop shown has sides *ab* and *cd* perpendicular to the figure plane, *bc* and *da* are parallel to it.

The total voltage will be a sum of voltages induced on each segment of the loop.



Voltage on each segment is:

$$e_{ind} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} \qquad (5.5.1)$$



The simplest DC machine

1) *ab*: In this segment, the velocity of the wire is tangential to the path of rotation. Under the pole face, velocity *v* is perpendicular to the magnetic field *B*, and the vector product *v* x *B* points into the page. Therefore, the voltage is

$$e_{ba} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} = \begin{cases} vBl & -\text{into page} - \text{under the pole face} \\ 0 & -\text{beyond the pole edges} \end{cases} \quad (5.6.1)$$

2) *bc*: In this segment, vector product *v* x *B* is perpendicular to *l*. Therefore, the voltage is zero.

3) *cd*: In this segment, the velocity of the wire is tangential to the path of rotation. Under the pole face, velocity *v* is perpendicular to the magnetic flux density *B*, and the vector product *v* x *B* points out of the page. Therefore, the voltage is

$$e_{dc} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} = \begin{cases} vBl & -\text{out of page} - \text{under the pole face} \\ 0 & -\text{beyond the pole edges} \end{cases} \quad (5.6.2)$$

4) *da*: In this segment, vector product *v* x *B* is perpendicular to *l*. Therefore, the voltage is zero.





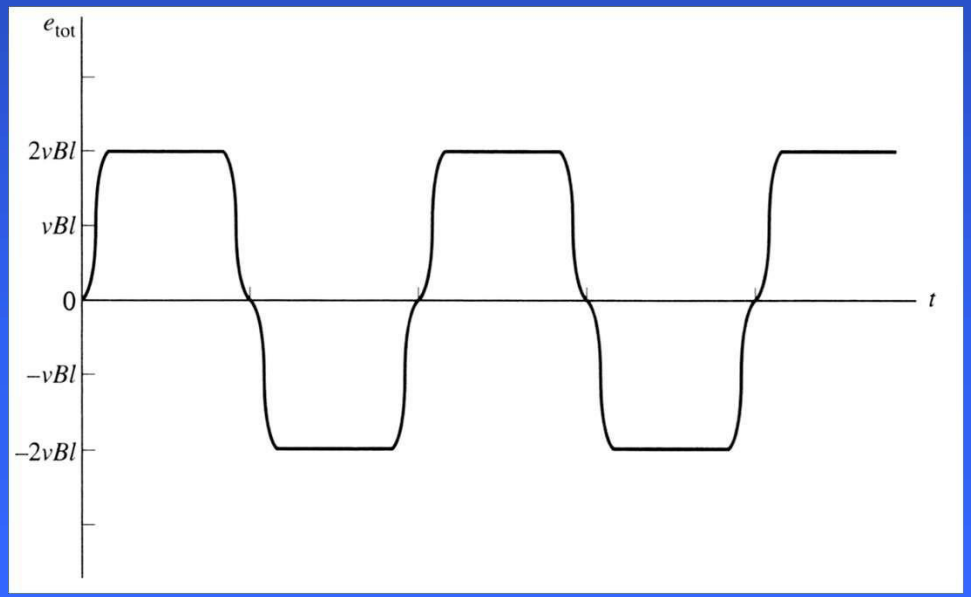
The simplest DC machine

The total induced voltage on the loop is:

$$e_{tot} = e_{ba} + e_{cb} + e_{dc} + e_{ad} \tag{5.7.1}$$

$$e_{tot} = \begin{cases} 2vBl & \textit{under the pole faces} \\ 0 & \textit{beyond the pole edges} \end{cases} \tag{5.7.2}$$

When the loop rotates through 180^0, segment *ab* is under the north pole face instead of the south pole face. Therefore, the direction of the voltage on the segment reverses but its magnitude reminds constant, leading to the total induced voltage to be



The simplest DC machine

The tangential velocity of the loop's edges is

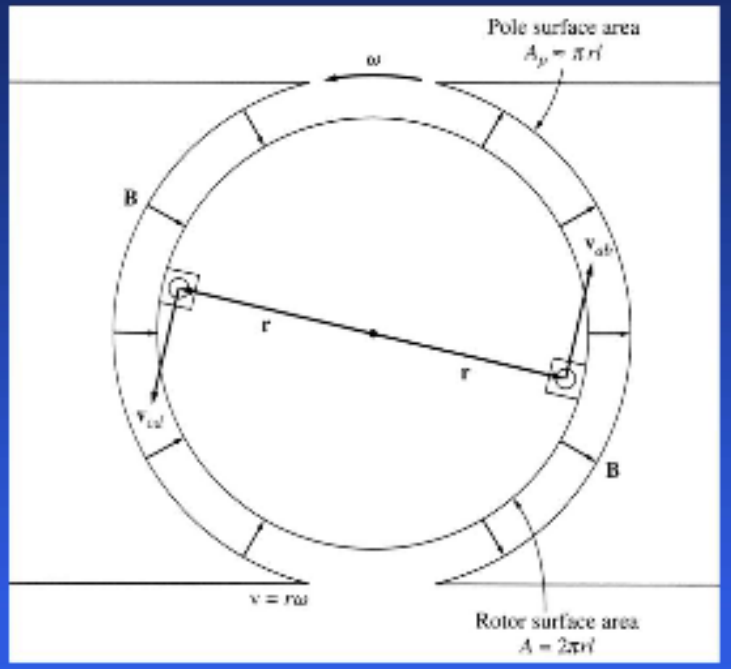
$$v = r\omega \tag{5.8.1}$$

where *r* is the radius from the axis of rotation to the edge of the loop. The total induced voltage:

$$e_{tot} = \begin{cases} 2r\omega Bl & \textit{under the pole faces} \\ 0 & \textit{beyond the pole edges} \end{cases} \tag{5.8.2}$$

The rotor is a cylinder with surface area $2\pi rl$. Since there are two poles, the area of the rotor under each pole is $A_p = \pi rl$. Therefore:

$$e_{tot} = \begin{cases} \dfrac{2}{\pi} A_p B\omega & \textit{under the pole faces} \\ 0 & \textit{beyond the pole edges} \end{cases} \tag{5.8.3}$$



The simplest DC machine

Assuming that the flux density *B* is constant everywhere in the air gap under the pole faces, the total flux under each pole is

$$\phi = A_p B \tag{5.9.1}$$

The total voltage is

$$e_{tot} = \begin{cases} \frac{2}{\pi}\phi\omega & \textit{under the pole faces} \\ 0 & \textit{beyond the pole edges} \end{cases} \tag{5.9.2}$$

The voltage generated in any real machine depends on the following factors:

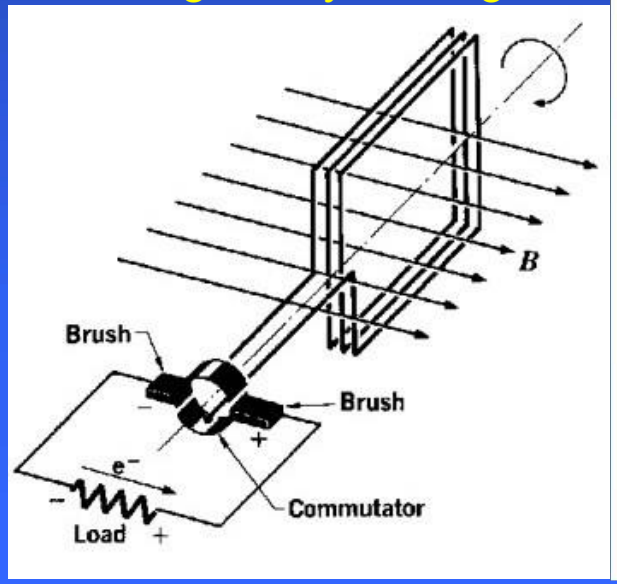
1. The flux inside the machine;
2. The rotation speed of the machine;
3. A constant representing the construction of the machine.

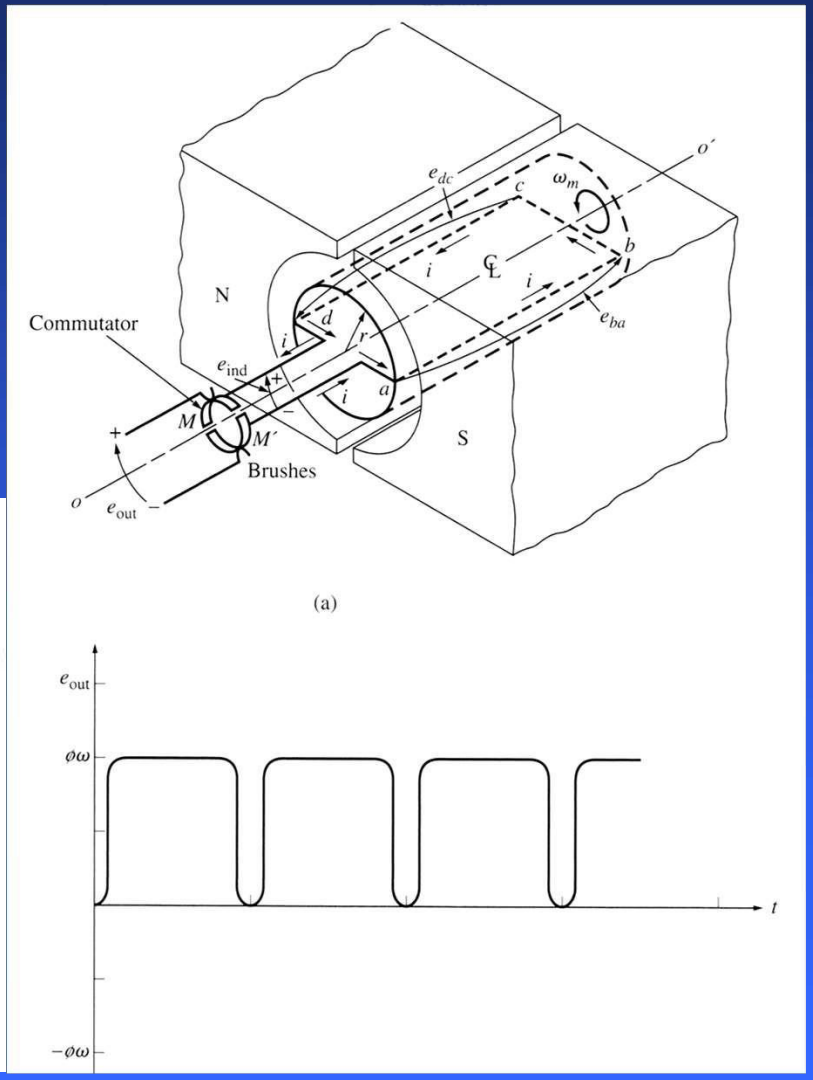
The simplest DC machine

2. Getting DC voltage out of a rotating loop

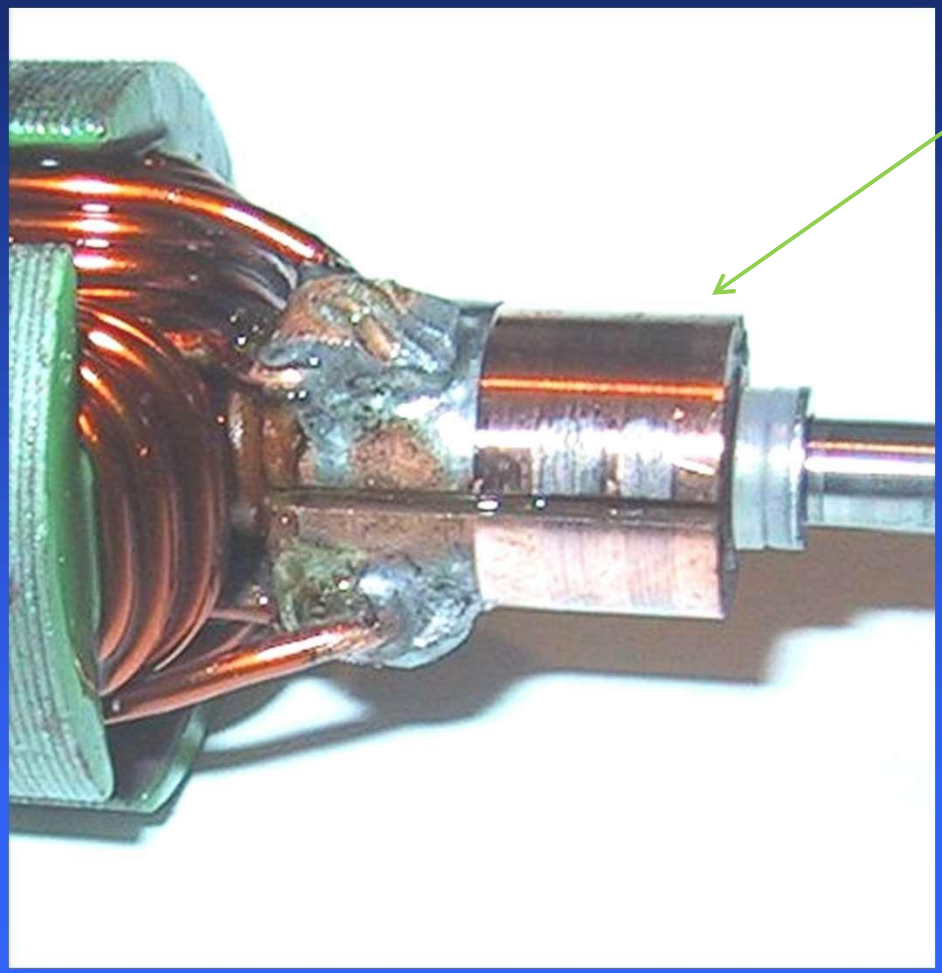
A voltage out of the loop is alternatively a constant positive value and a constant negative value.

One possible way to convert an alternating voltage to a constant voltage is by adding a commutator segment/brush circuitry to the end of the loop. Every time the voltage of the loop switches direction, contacts switch connection.





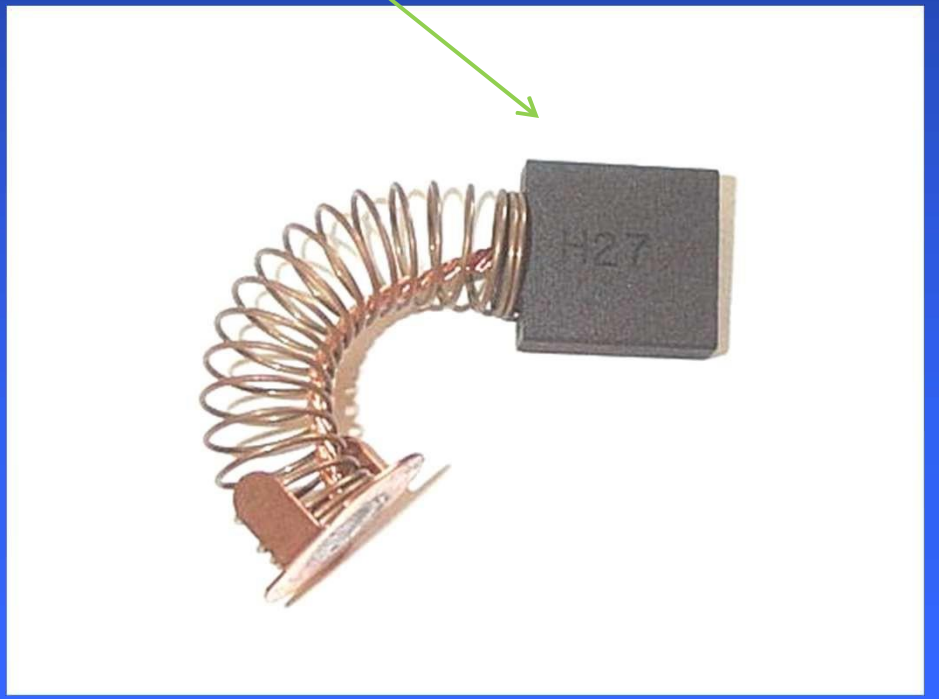
The simplest DC machine



segments

brushes





The simplest DC machine

3. The induced torque in the rotating loop

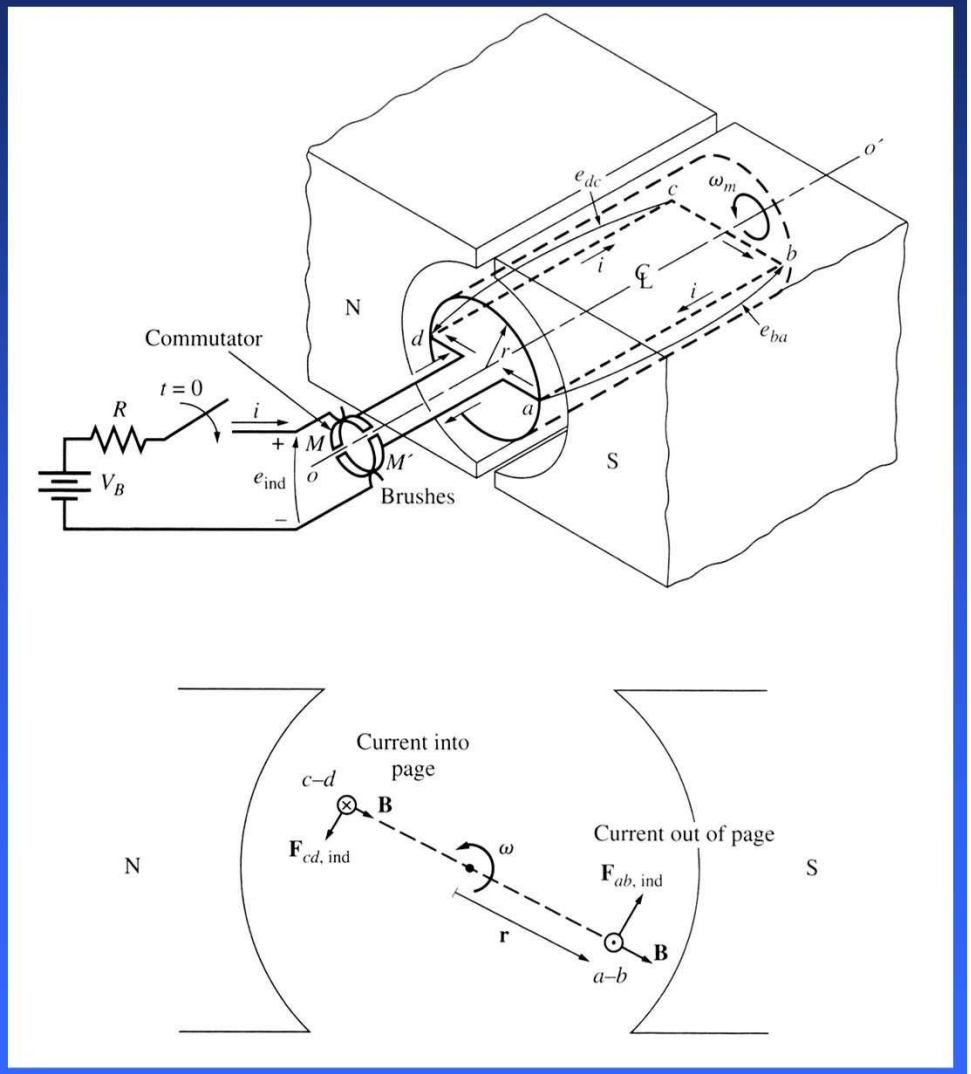
Assuming that a battery is connected to the DC machine, the force on a segment of a loop is:

$$F = i\left(\mathbf{l}\times\mathbf{B}\right) \qquad (5.12.1)$$

And the torque on the segment is

$$\tau = rF\sin\theta \qquad (5.12.2)$$

Where θ is the angle between r and F. Therefore, the torque is zero when the loop is beyond the pole edges.



The simplest DC machine

When the loop is under the pole faces:

1. Segment *ab*:

$$F_{ab} = i(\mathbf{l}\times\mathbf{B}) = ilB \tag{5.13.1}$$

$$\tau_{ab} = rF\sin\theta = r(ilB)\sin 90^\circ = rilB \quad ccw \tag{5.13.2}$$

2. Segment *bc*:

$$F_{ab} = i(\mathbf{l}\times\mathbf{B}) = 0 \tag{5.13.3}$$

$$\tau_{ab} = rF\sin\theta = 0 \tag{5.13.4}$$

3. Segment *cd*:

$$F_{ab} = i(\mathbf{l}\times\mathbf{B}) = ilB \tag{5.13.5}$$

$$\tau_{ab} = rF\sin\theta = r(ilB)\sin 90^\circ = rilB \quad ccw \tag{5.13.6}$$

4. Segment *da*:

$$F_{ab} = i(\mathbf{l}\times\mathbf{B}) = 0 \tag{5.13.7}$$

$$\tau_{ab} = rF\sin\theta = 0 \tag{5.13.8}$$





The simplest DC machine

The resulting total induced torque is

$$\tau_{ind} = \tau_{ab} + \tau_{bc} + \tau_{cd} + \tau_{da} \tag{5.14.1}$$

$$\tau_{ind} = \begin{cases} 2rilB & \text{under the pole faces} \\ 0 & \text{beyond the pole edges} \end{cases} \tag{5.14.2}$$

Since $A_p \cong \pi r l$ and $\phi = A_p B$ (5.14.3)

$$\tau_{ind} = \begin{cases} \frac{2}{\pi}\phi i & \text{under the pole faces} \\ 0 & \text{beyond the pole edges} \end{cases} \tag{5.14.4}$$

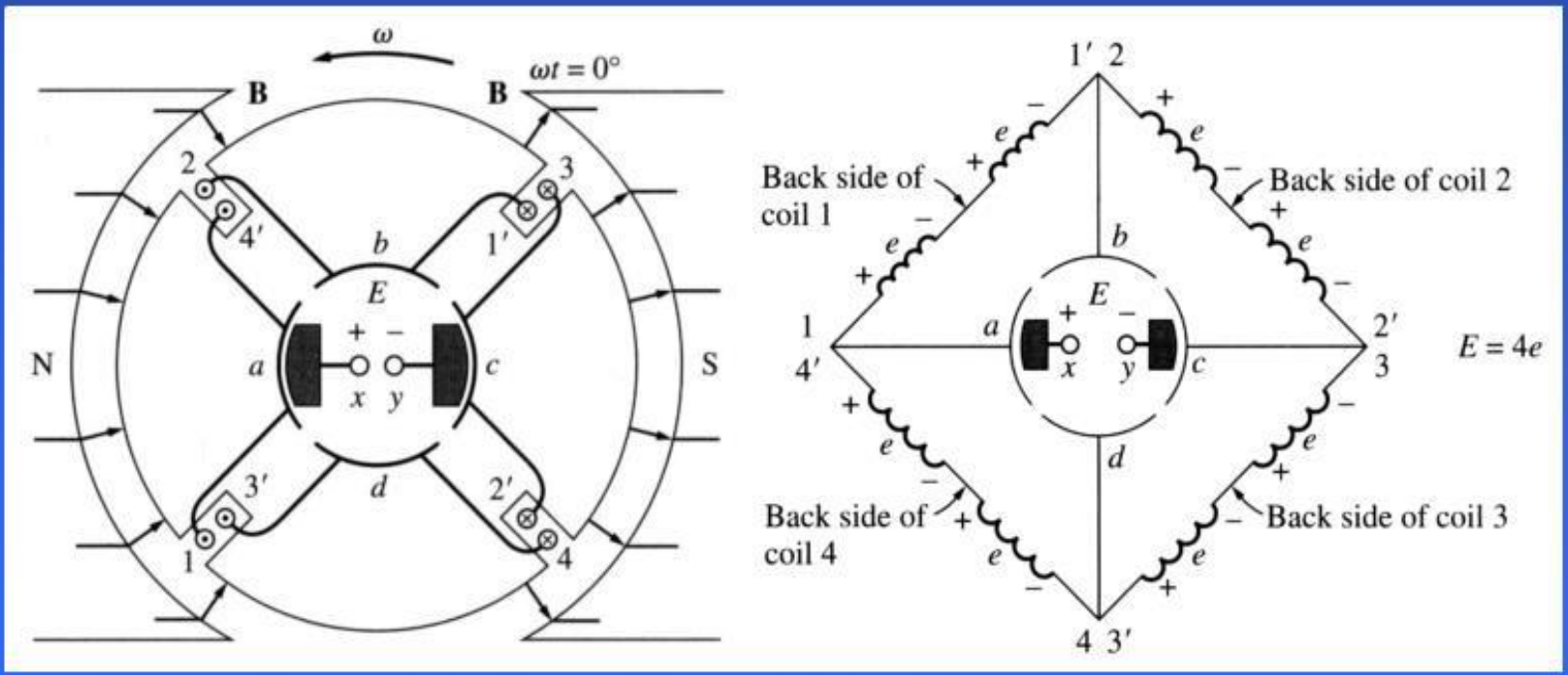
The torque in any real machine depends on the following factors:

1. The flux inside the machine;
2. The current in the machine;
3. A constant representing the construction of the machine.

Commutation in a simple 4-loop DC machine

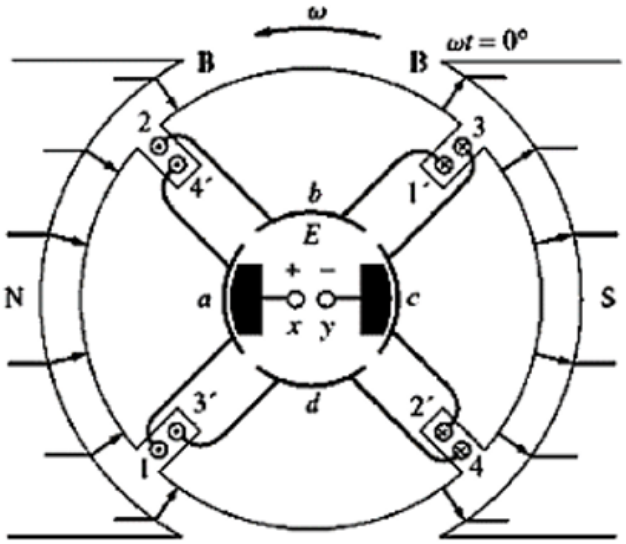
Commutation is the process of converting the AC voltages and currents in the rotor of a DC machine to DC voltages and currents at its terminals.

A simple 4-loop DC machine has four complete loops buried in slots curved in the laminated steel of its rotor. The pole faces are curved to make a uniform air-gap. The four loops are laid into the slots in a special manner: the innermost wire in each slot (end of each loop opposite to the "unprimed") is indicated by a prime.



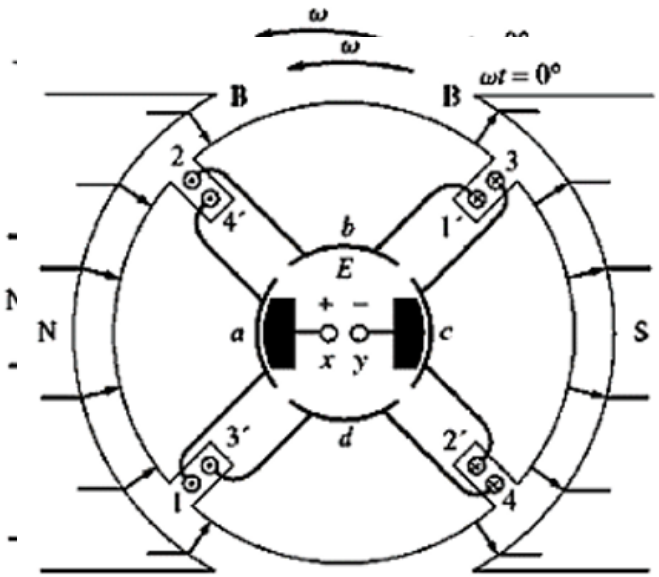
Loop 1 stretches between commutator segments *a* and *b*, loop 2 stretches between segments *b* and *c*…

Describing the rotor(armature) winding arrangement of the DC machine



- The armature has 4 coils (1-1′,2-2′,3-3′,4-4′)
- Each coil has 1 turn
- There are a total of 8 conductors (4x2)
- There are 2 brushes (x,y)
- There are 4 commutator segments (a,b,c,d)
- There are 4 slots (1,2,3,4)
- The armature has **progressive lap winding**

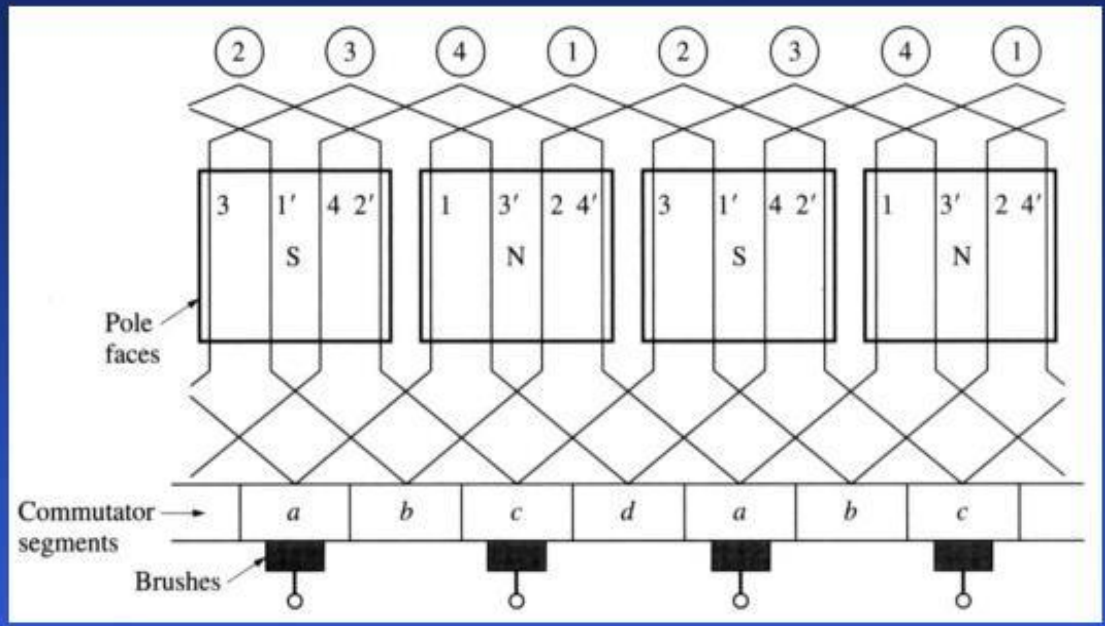
Describing the rotor(armature) winding arrangement of the DC machine



- The armature has 4 coils (1-1’,2-2’,3-3’,4-4’)
- Each coil has 1 turn
- There are a total of 8 conductors (4x2)
- There are 2 brushes (x,y)
- There are 4 commutator segments (a,b,c,d)
- There are 4 slots (1,2,3,4)
- The armature has **progressive lap winding**

Commutation in a simple 4-loop DC machine

At a certain time instance, when $\omega t = 0^0$, the 1, 2, 3', and 4' ends of the loops are under the north pole face and the 1', 2', 3, and 4 ends of the loops are under the south pole face. The voltage in each of 1, 2, 3', and 4' ends is given by



$$e_{ind} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} = vBl \qquad (5.16.1)$$

– positive, out of the page

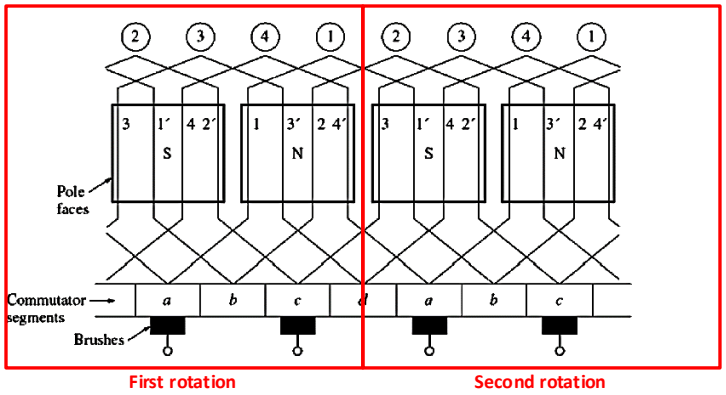
The voltage in each of 1', 2', 3, and 4 ends is

$$e_{ind} = (\mathbf{v} \times \mathbf{B}) \times \mathbf{l} = vBl \quad \text{– positive, into the page} \qquad (5.16.2)$$

If the induced voltage on any side of a loop is (5.16.1), the total voltage at the brushes of the DC machine is

$$E = 4e \qquad at\ \omega t = 0° \qquad (5.16.3)$$

Commutation in a Simple Four-Loop DC Machine



Winding diagram of this machine showing the interconnections of the rotor loops at time wt=0°.

Commutation in a Simple Four-Loop DC Machine

At the instant shown in Figure 8–7, the 1, 2, 3′, and 4′ ends of the loops are under the north pole face, while the 1′, 2′, 3, and 4 ends of the loops are under the south pole face. The voltage in each of the 1, 2, 3′, and 4′ ends of the loops is given by

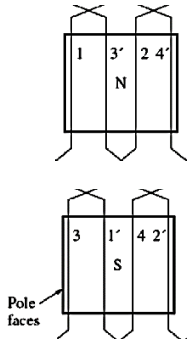
$$e_{ind} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} \qquad (1\text{–}45)$$

$$e_{ind} = vBl \quad \text{positive out of page} \qquad (8\text{–}17)$$

The voltage in each of the 1′, 2′, 3, and 4 en … en by

$$e_{ind} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} \qquad (1\text{–}45)$$

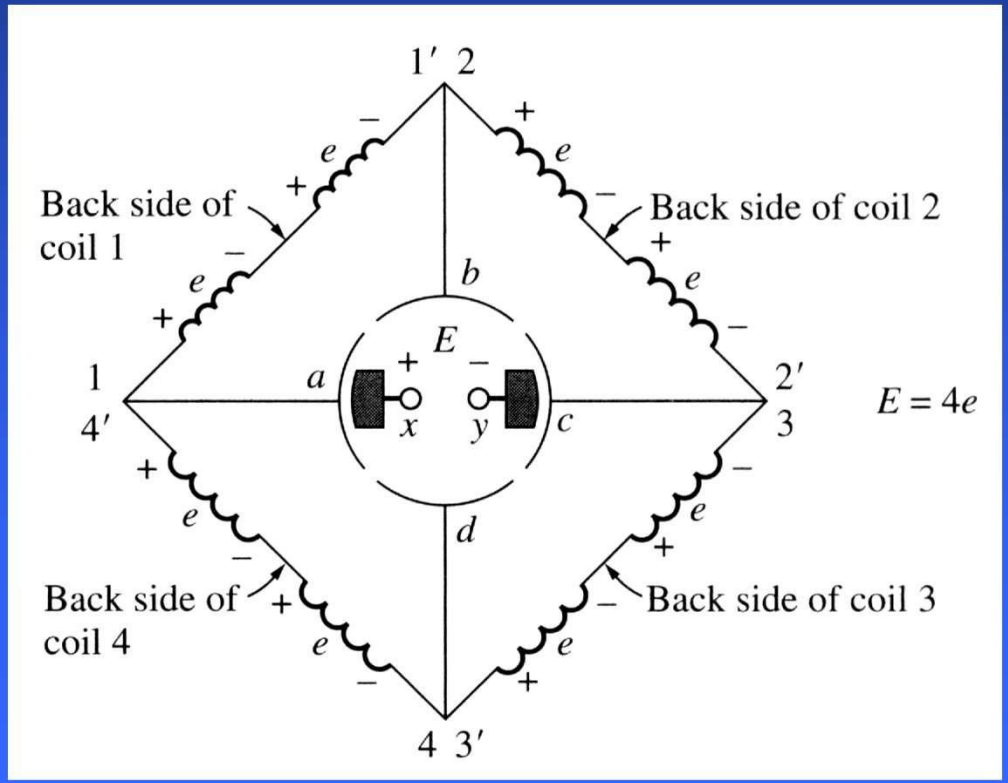
$$= vBl \quad \text{positive into the page} \qquad (8\text{–}18)$$



Check the polarities using slides 8 and 10 !

Commutation in a simple 4-loop DC machine

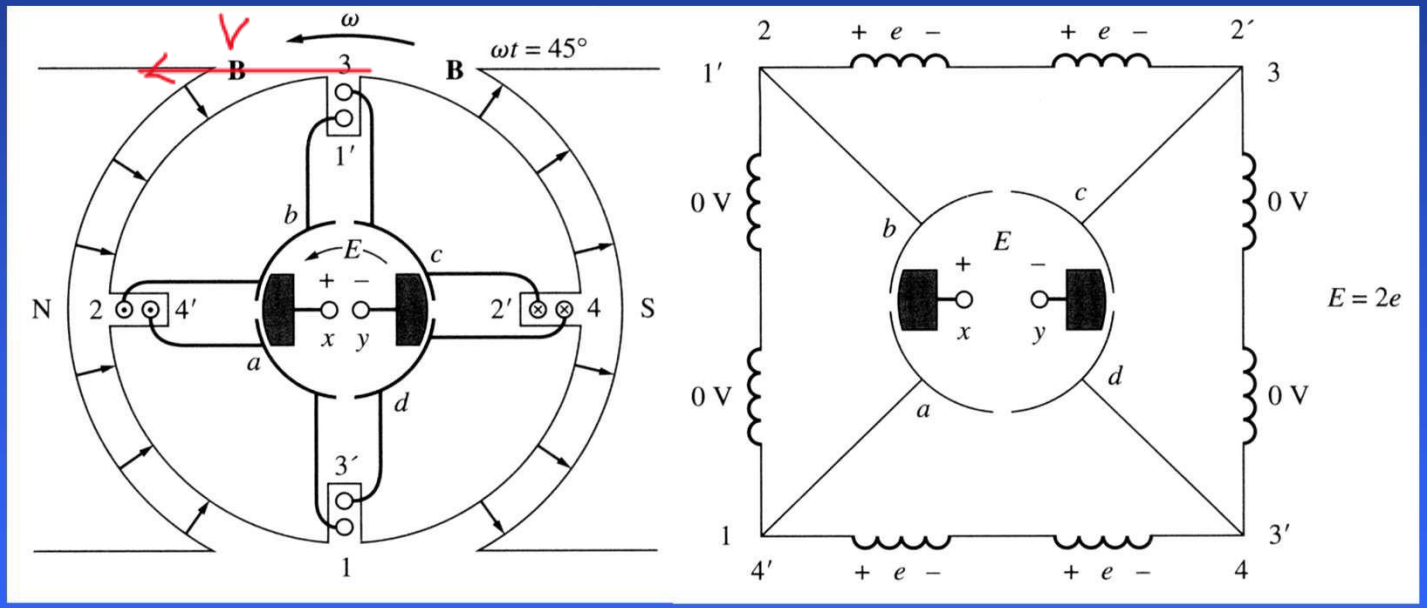
We notice that there are two parallel paths for current through the machine! The existence of two or more parallel paths for rotor current is a common feature of all commutation schemes.



Commutation in a simple 4-loop DC machine

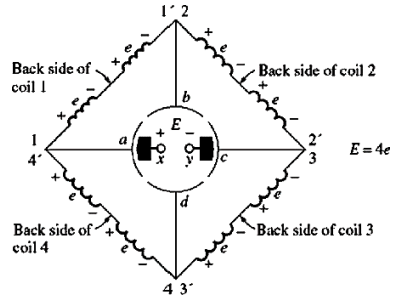
If the machine keeps rotating, at $\omega t = 45^0$, loops 1 and 3 have rotated into the gap between poles, so the voltage across each of them is zero. At the same time, the brushes short out the commutator segments *ab* and *cd*.

This is ok since the voltage across loops 1 and 3 is zero and only loops 2 and 4 are under the pole faces. Therefore, the total terminal voltage is



$$E = 2e \qquad at\ \omega t = 45^\circ \tag{5.18.1}$$

Commutation in a Simple Four-Loop DC Machine



The voltages on the rotor conductors at wt=0°.

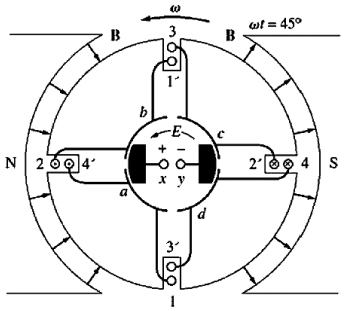
- Notice that there are **two parallel paths** for current in the armature.
- The existence of two or more parallel paths for rotor current is a common feature of all commutation schemes.

The total voltage at the brushes of the machine is

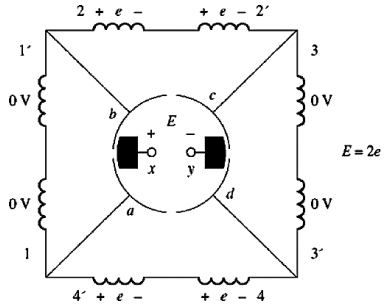
$$E = 4e \qquad \omega t = 0°$$

Commutation in a Simple Four-Loop DC Machine

What happens to the voltage of the terminals as the rotor **continues to rotate**?

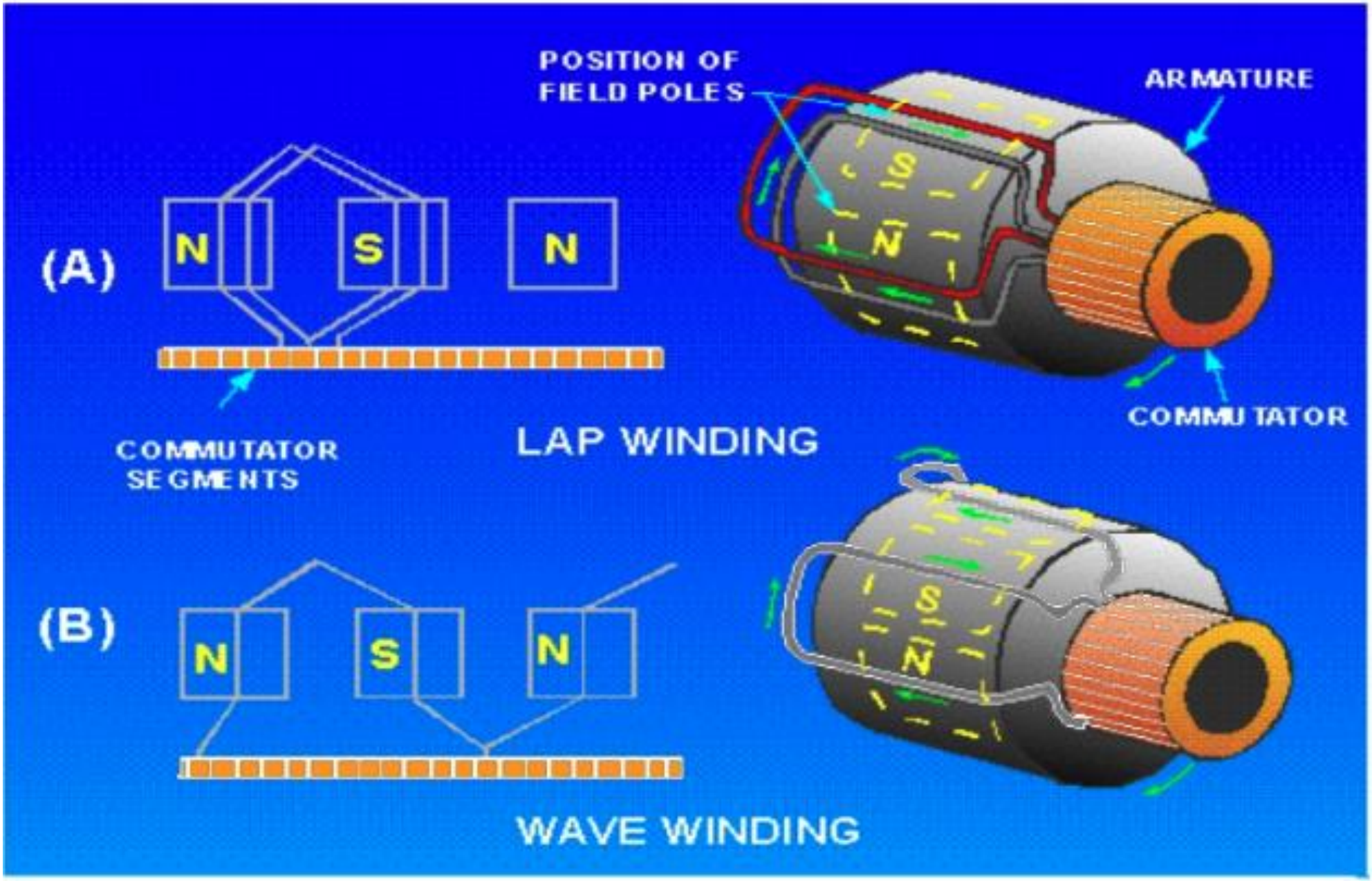


A four-loop two-pole dc machine shown at time wt=45°

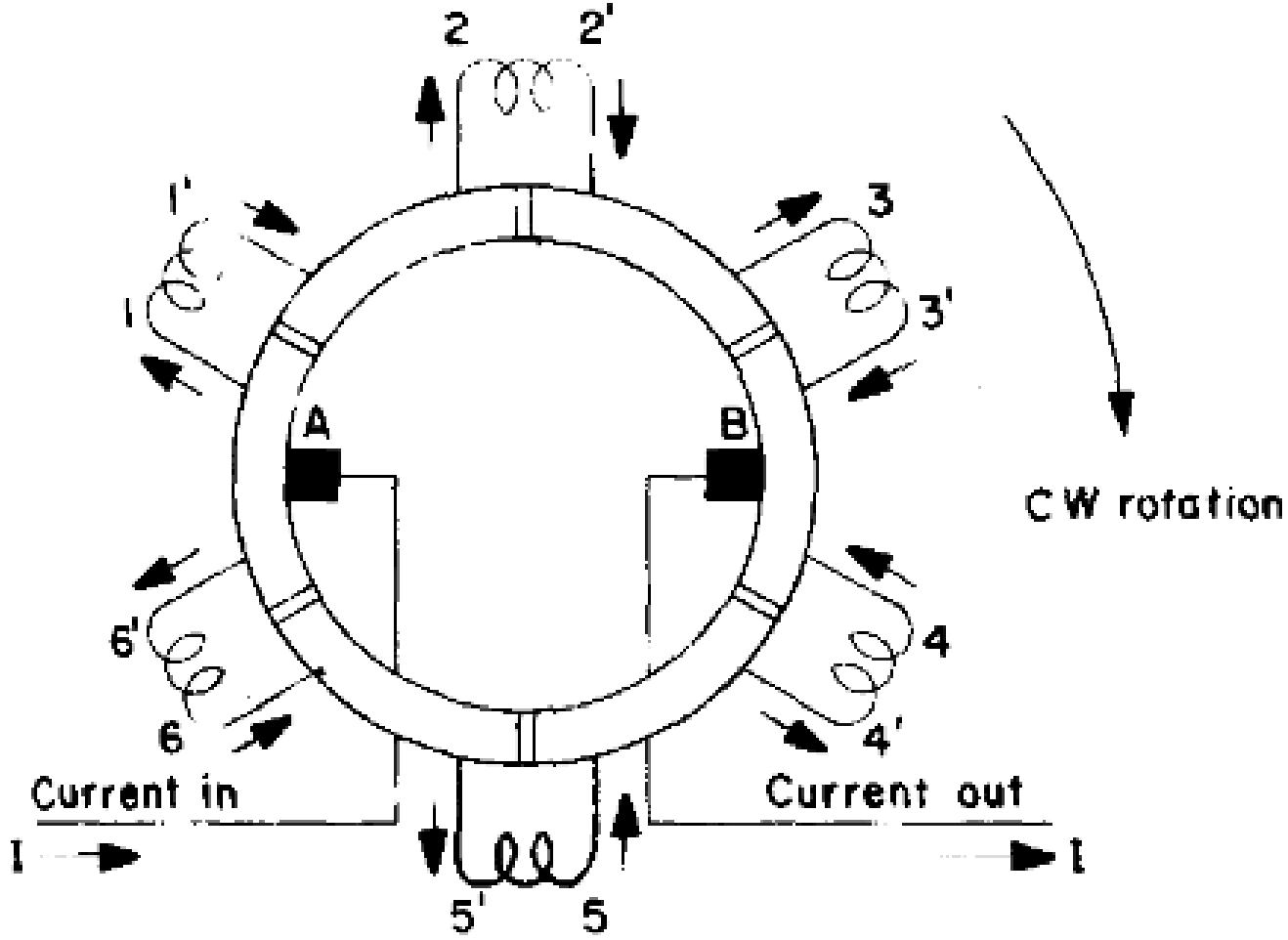


The voltages on the rotor conductors at wt=45°.

The total voltage at the brushes of the machine is → $E = 2e \qquad \omega t = 45°$

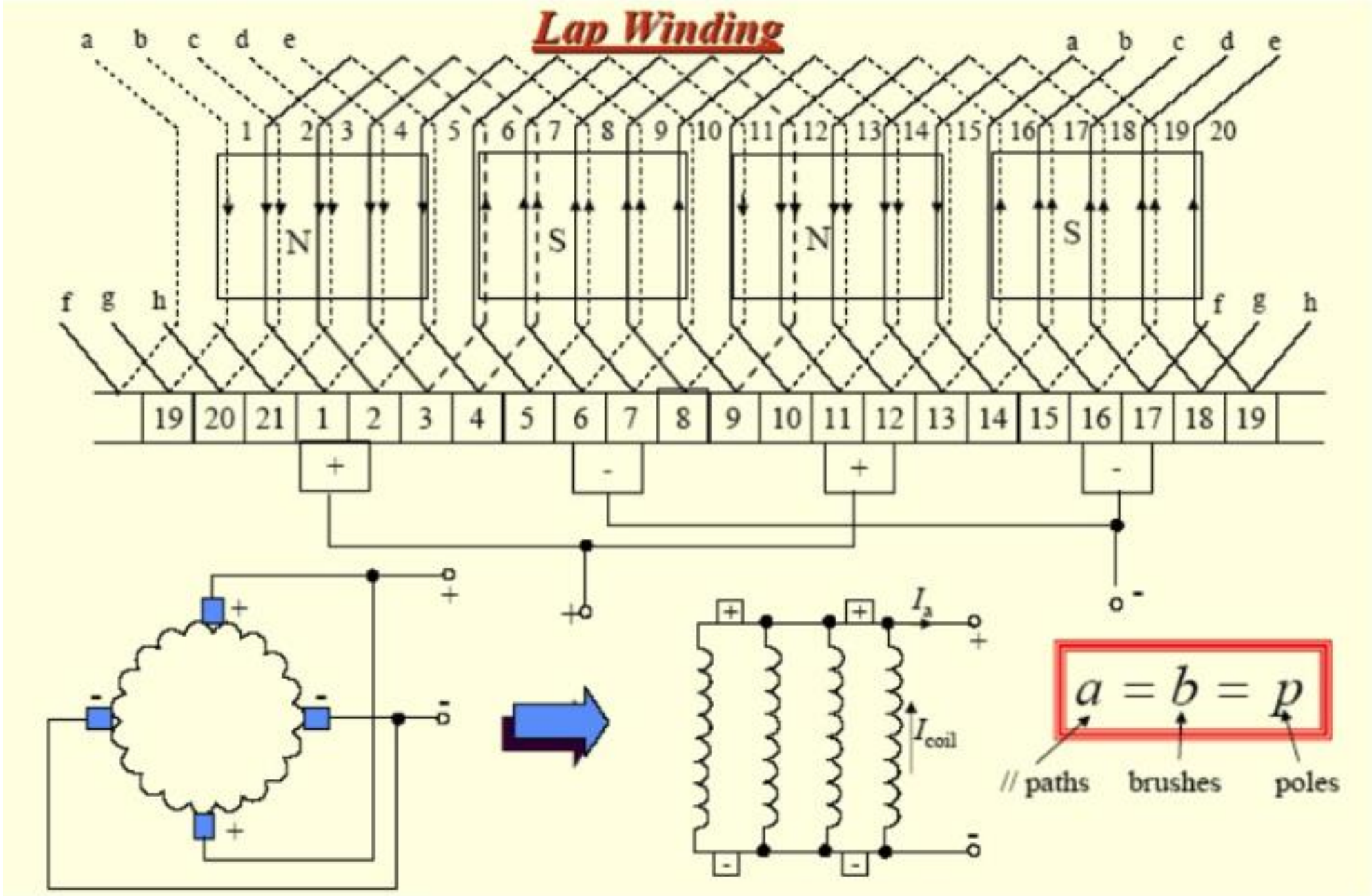

POSITION OF
FIELD POLES
ARMATURE
(A)
N
S
N
S
N
COMMUTATOR
COMMUTATOR
SEGMENTS
LAP WINDING
(B)
N
S
N
S
N
WAVE WINDING

A representation of current flow through a motor armature

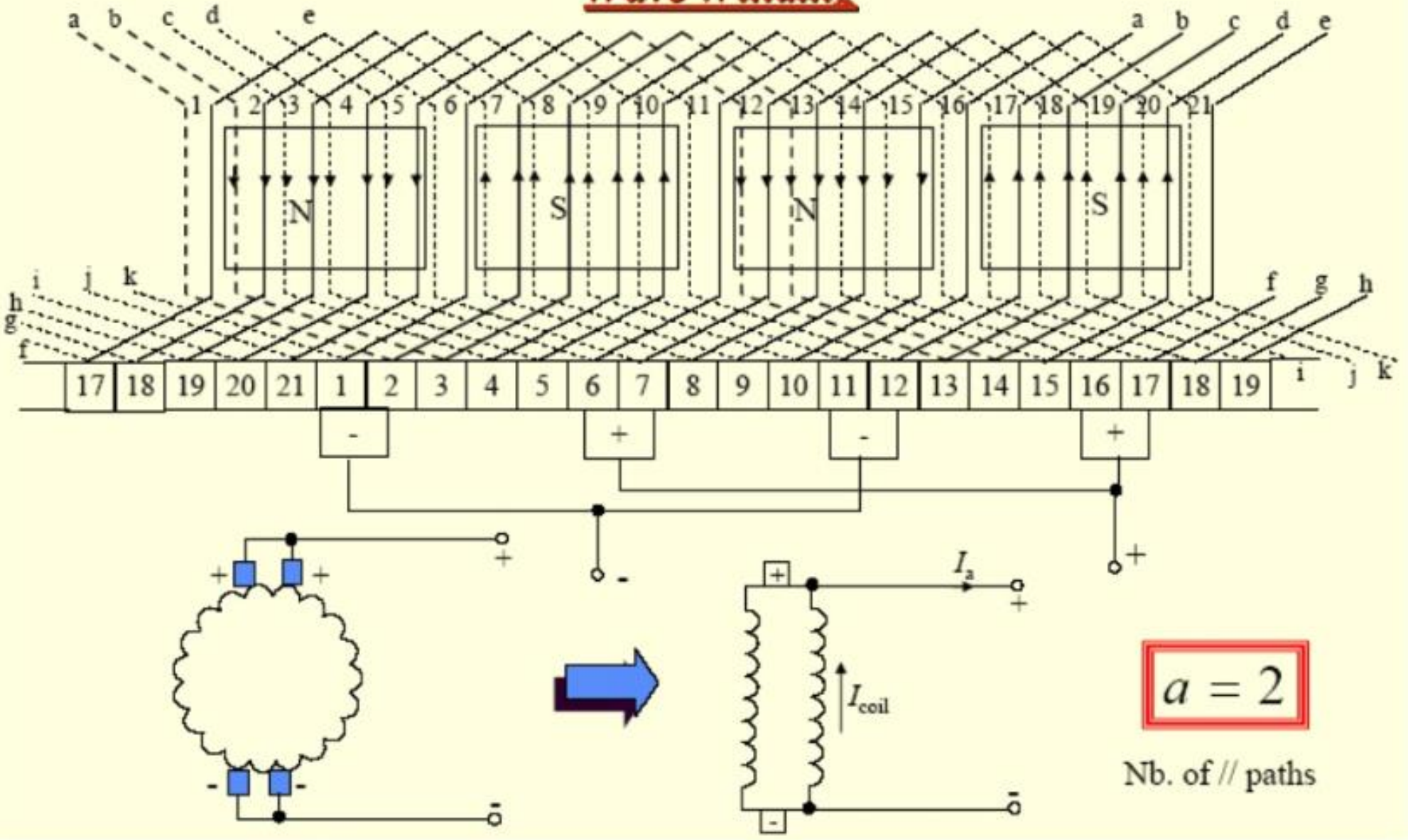


If the current is reversed all symmetric poles are replaced with each other. So the machine rotation is also reversed.

Lap Winding


a b c d e
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
N
S
N
S
f g h
19 20 21 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19
+
-
+
-
I_a
I_{coil}
$a = b = p$
// paths
brushes
poles

Wave Winding

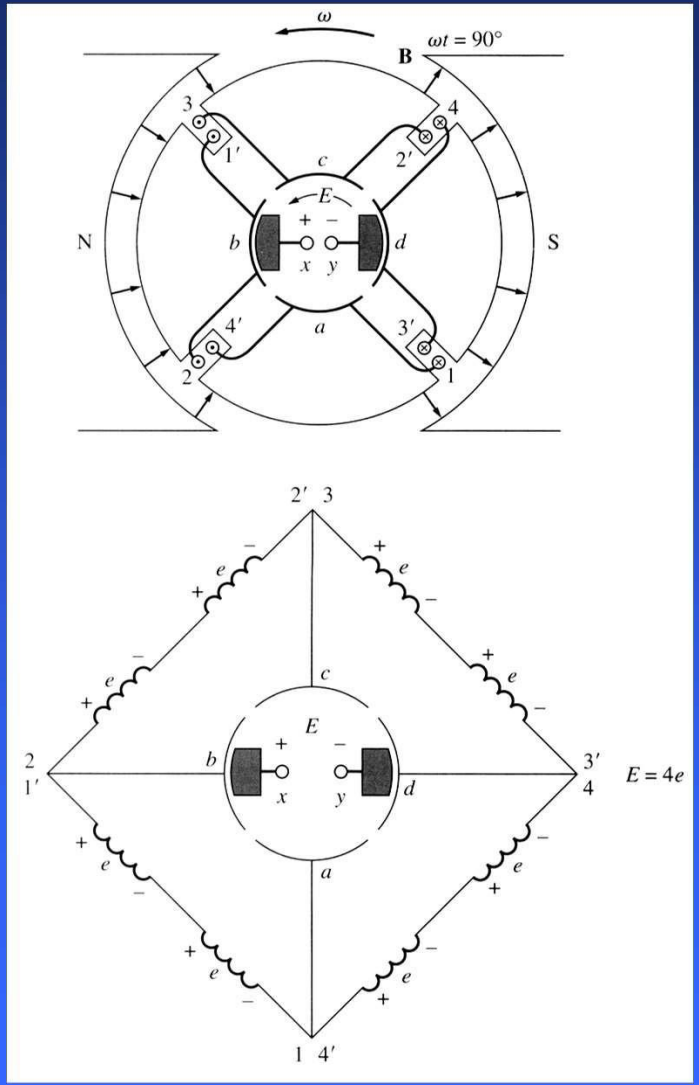


Commutation in a simple 4-loop DC machine

At $\omega t = 90^0$, the loop ends 1', 2, 3, and 4' are under the north pole face, and the loop ends 1, 2', 3', and 4 are under the south pole face. The voltages are built up out of page for the ends under the north pole face and into the page for the ends under the south pole face. Four voltage-carrying ends in each parallel path through the machine lead to the terminal voltage of

$$E = 4e \qquad at\ \omega t = 90° \tag{5.16.3}$$

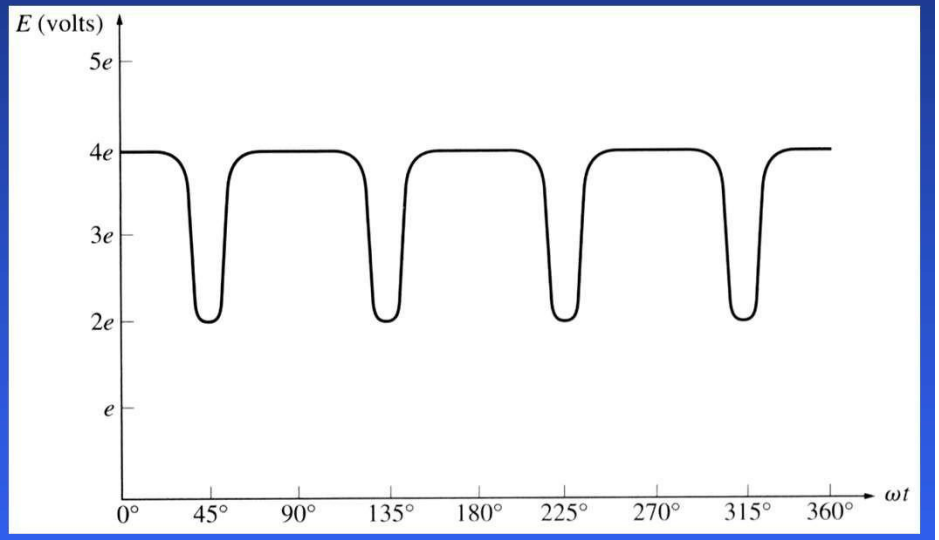
We notice that the voltages in loops 1 and 3 have reversed compared to $\omega t = 0^0$. However, the loops' connections have also reversed, making the total voltage being of the same polarity.



Commutation in a simple 4-loop DC machine

When the voltage reverses in a loop, the connections of the loop are also switched to keep the polarity of the terminal voltage the same.

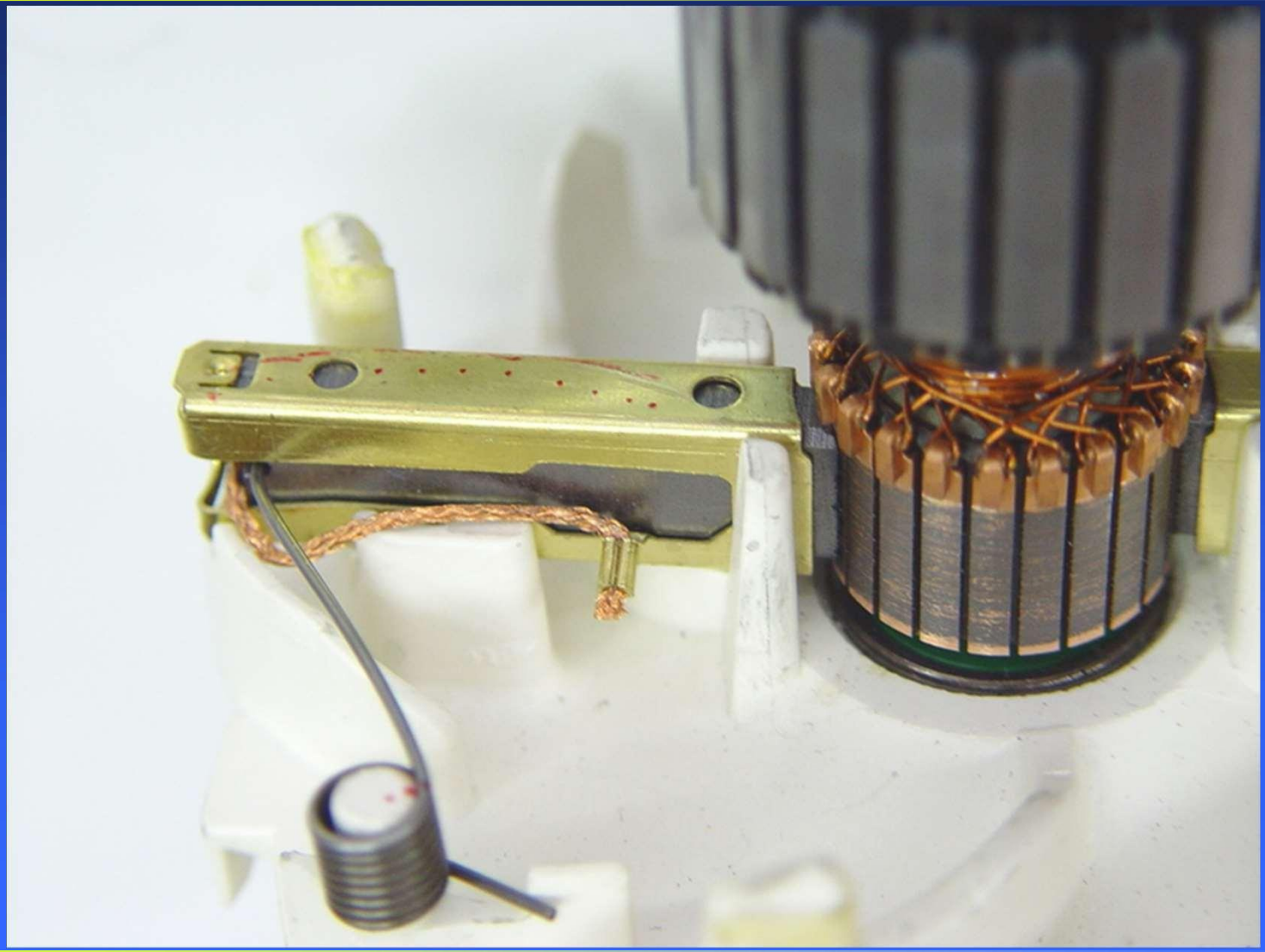
The terminal voltage of this 4-loop DC machine is still not constant over time, although it is a better approximation to a constant DC level than what is produced by a single rotating loop.



Increasing the number of loops on the rotor, we improve our approximation to perfect DC voltage.

Commutator segments are usually made out of copper bars and the brushes are made of a mixture containing graphite to minimize friction between segments and brushes.

Example of a commutator…



Problems with commutation in real DC machines

1. Armature reaction

If the magnetic field windings of a DC machine are connected to the power source and the rotor is turned by an external means, a voltage will be induced in the conductors of the rotor.

This voltage is rectified and can be supplied to external loads. However, if a load is connected, a current will flow through the armature winding.

This current produces its own magnetic field that distorts the original magnetic field from the machine's poles.

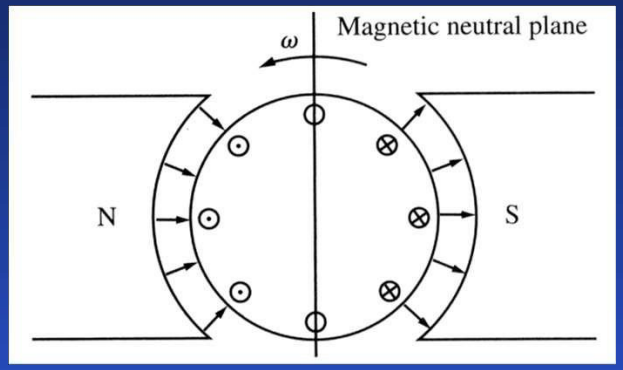
This distortion of the machine's flux as the load increases is called armature reaction and can cause two problems:

Problems with commutation in real DC machines

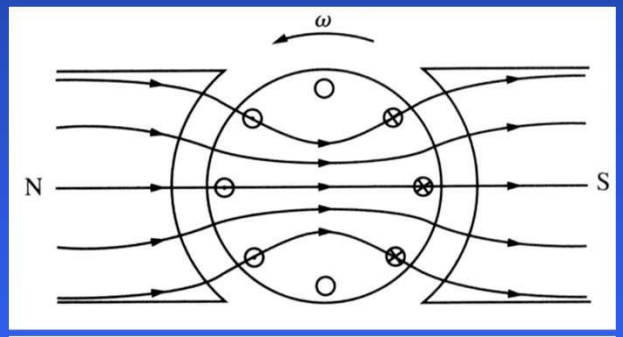
1. Armature reaction

1) Neutral-plane shift: The magnetic neutral plane is the plane within the machine where the velocity of the rotor wires is exactly parallel to the magnetic flux lines, so that the induced voltage in the conductors in the plane is exactly zero.

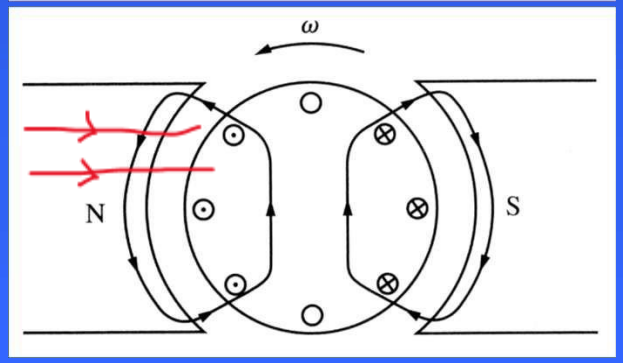
Problems with commutation in real DC machines



A two-pole DC machine: initially, the pole flux is uniformly distributed and the magnetic neutral plane is vertical.

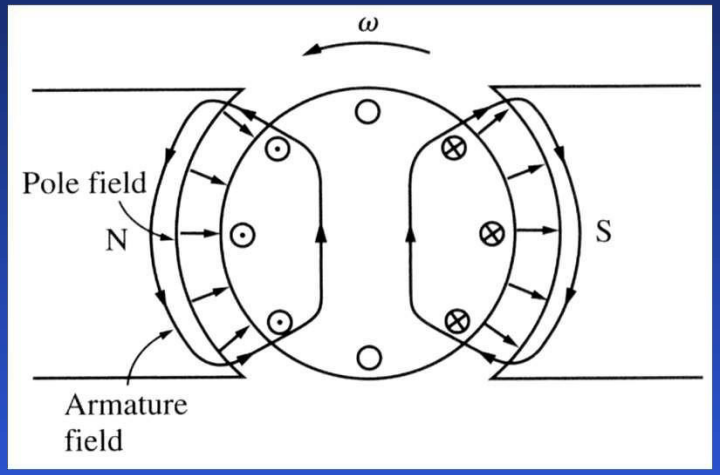


The effect of the air gap on the pole flux.

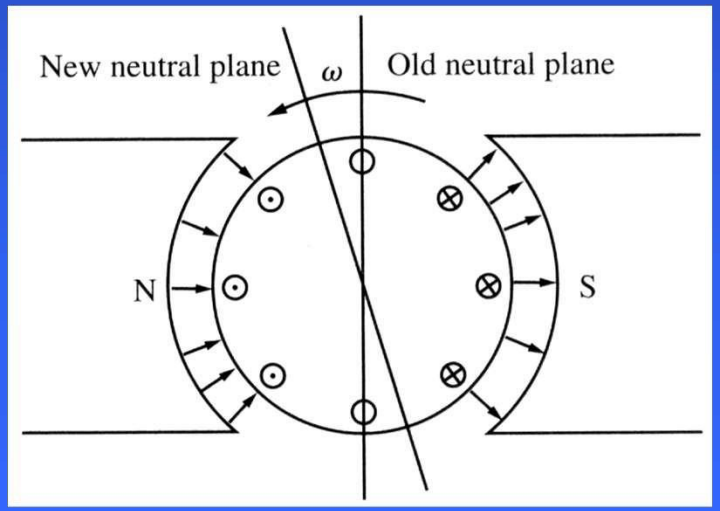


When the load is connected, a current – flowing through the rotor – will generate a magnetic field from the rotor windings.

Problems with commutation in real DC machines



This rotor magnetic field will affect the original magnetic field from the poles. In some places under the poles, both fields will sum together, in other places, they will subtract from each other



Therefore, the net magnetic field will not be uniform and the neutral plane will be shifted.

In general, the neutral plane shifts in the direction of motion for a generator and opposite to the direction of motion for a motor. The amount of the shift depends on the load of the machine.

Problems with commutation in real DC machines

The commutator must short out the commutator segments right at the moment when the voltage across them is zero. The neutral-plane shift may cause the brushes short out commutator segments with a non-zero voltage across them. This leads to arcing and sparkling at the brushes!









Problems with commutation in real DC machines

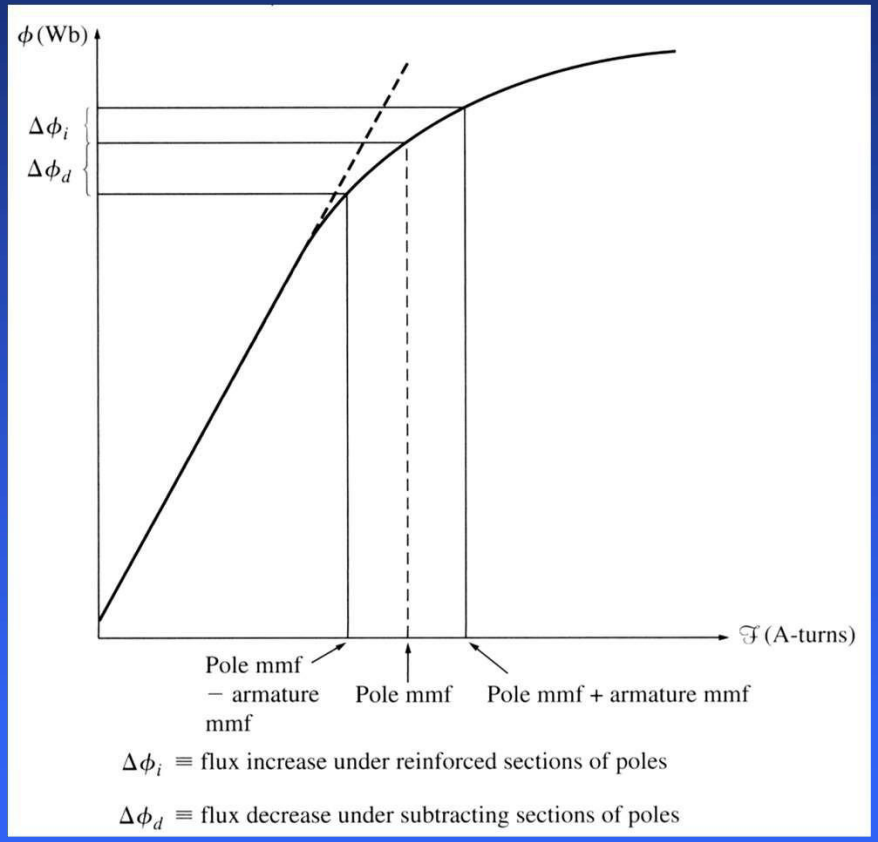
2) Flux weakening.

Most machines operate at flux densities near the saturation point.

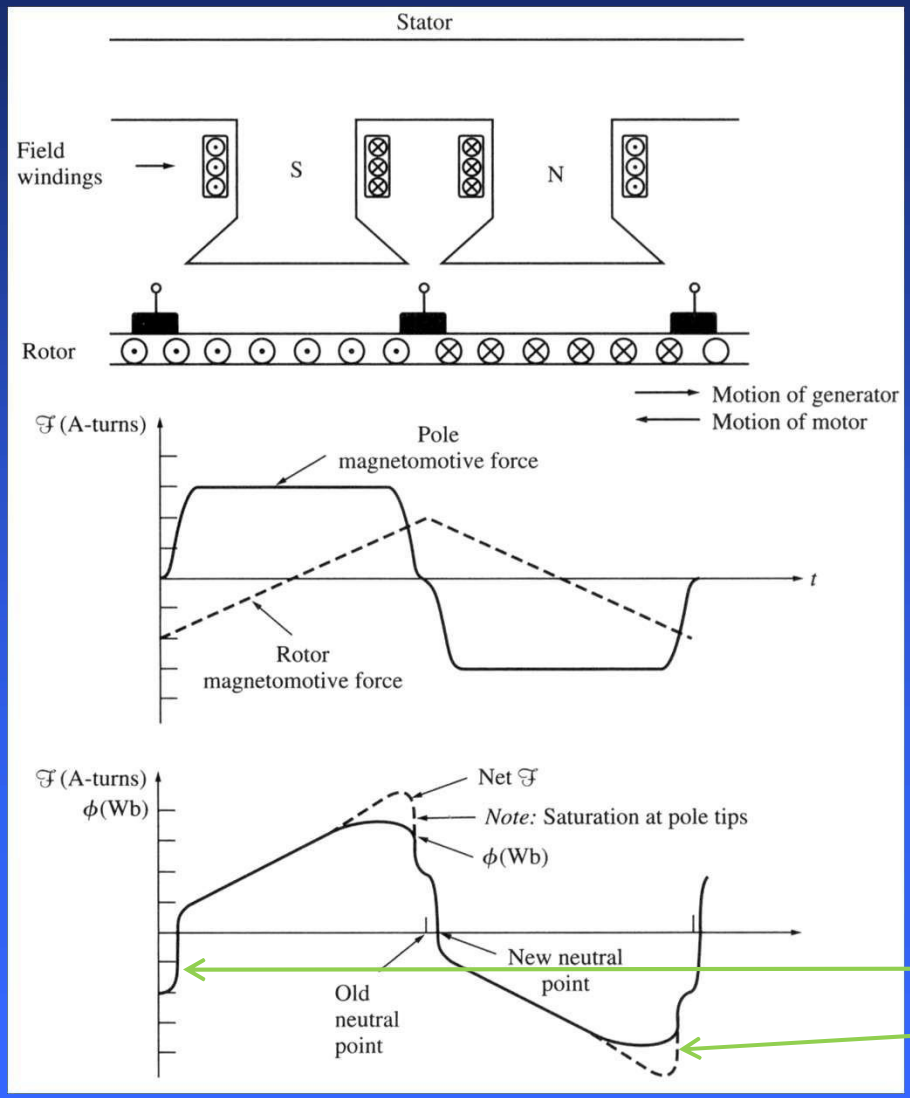
At the locations on the pole surfaces where the rotor mmf adds to the pole mmf, only a small increase in flux occurs (due to saturation).

However, at the locations on the pole surfaces where the rotor mmf subtracts from the pole mmf, there is a large decrease in flux.

Therefore, the total average flux under the entire pole face decreases.



Problems with commutation in real DC machines



In generators, flux weakening reduces the voltage supplied by a generator.

In motors, flux weakening leads to increase of the motor speed. Increase of speed may increase the load, which, in turns, results in more flux weakening. Some shunt DC motors may reach runaway conditions this way…

Observe a considerable decrease in the region where two mmfs are subtracted and a saturation…

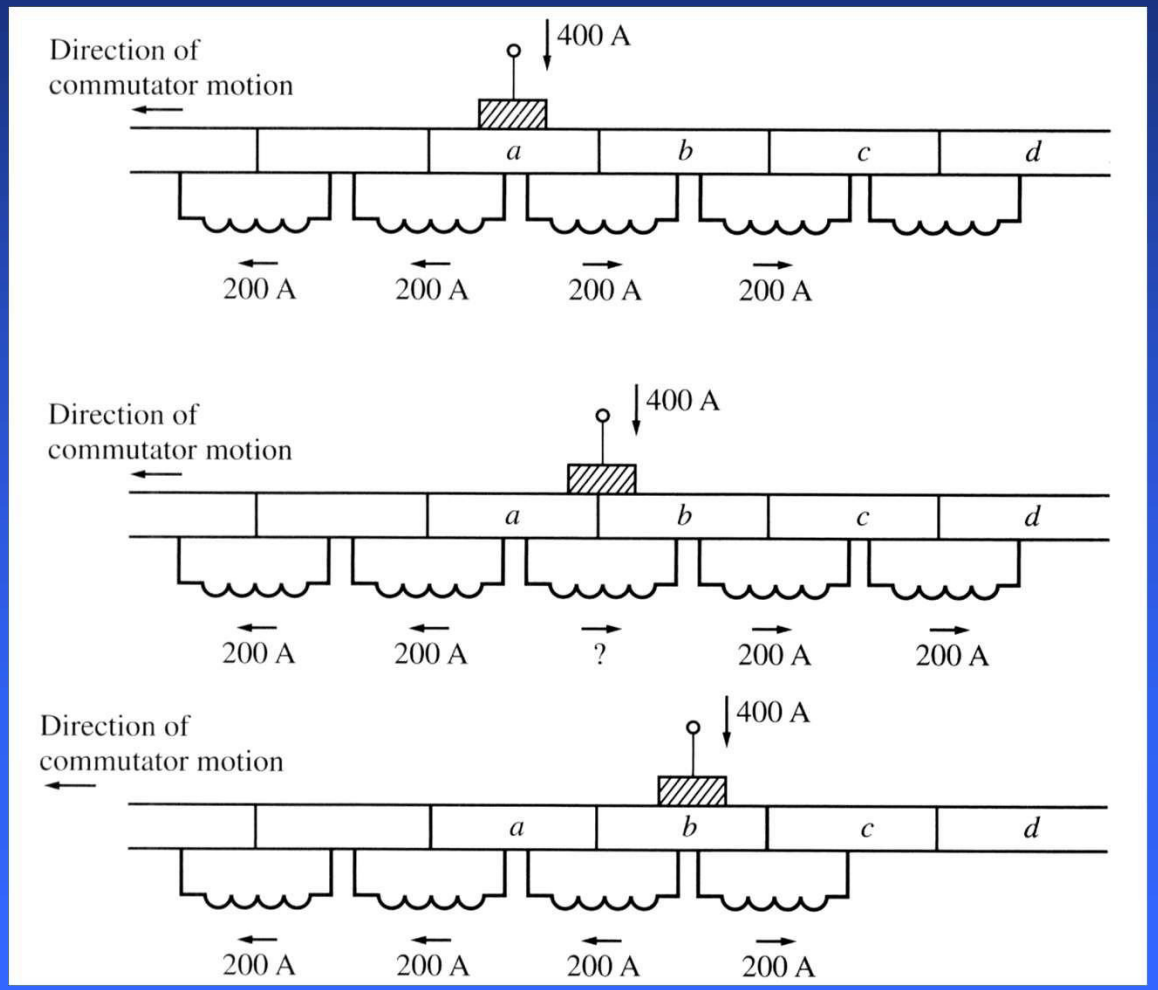
Problems with commutation in real DC machines

2. *L di/dt* voltages

This problem occurs in commutator segments being shorten by brushes and is called sometimes an inductive kick.

Assuming that the current in the brush is 400 A, the current in each path is 200 A. When a commutator segment is shorted out, the current flow through that segment must reverse.

Assuming that the machine is running at 800 rpm and has 50 commutator segments, each segment moves under the brush and clears it again in 0.0015 s.



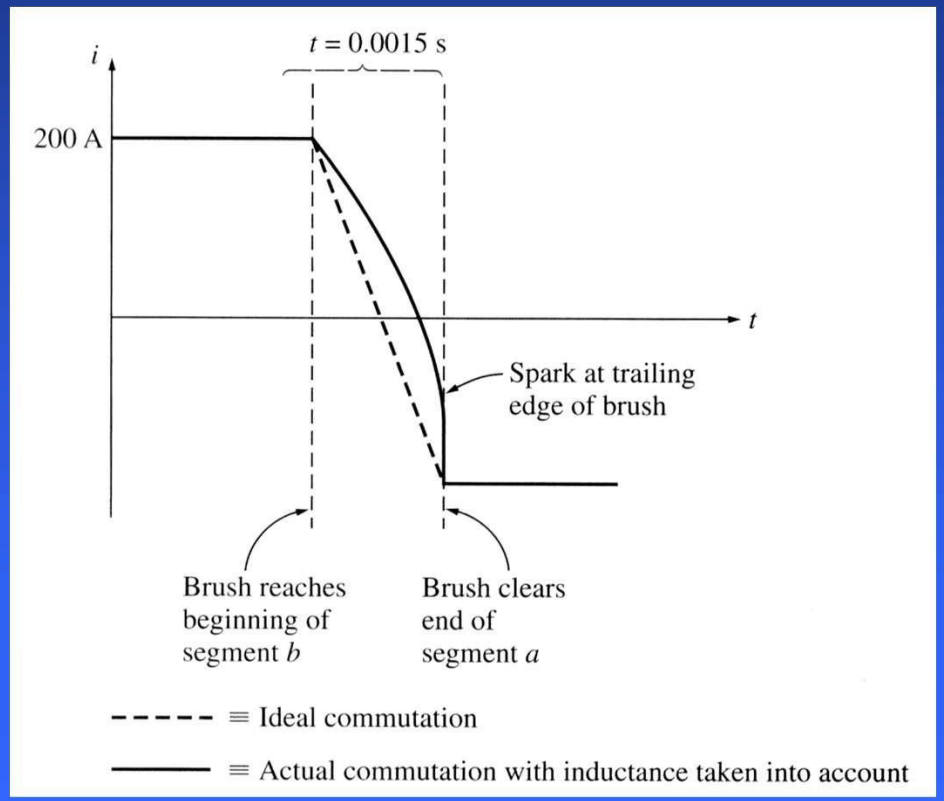
Problems with commutation in real DC machines

The rate of change in current in the shorted loop averages

$$\frac{di}{dt} = \frac{400}{0.0015} = 266667\ A/s$$

Therefore, even with a small inductance in the loop, a very large inductive voltage kick *L di/dt* will be induced in the shorted commutator segment.

This voltage causes sparkling at the brushes.

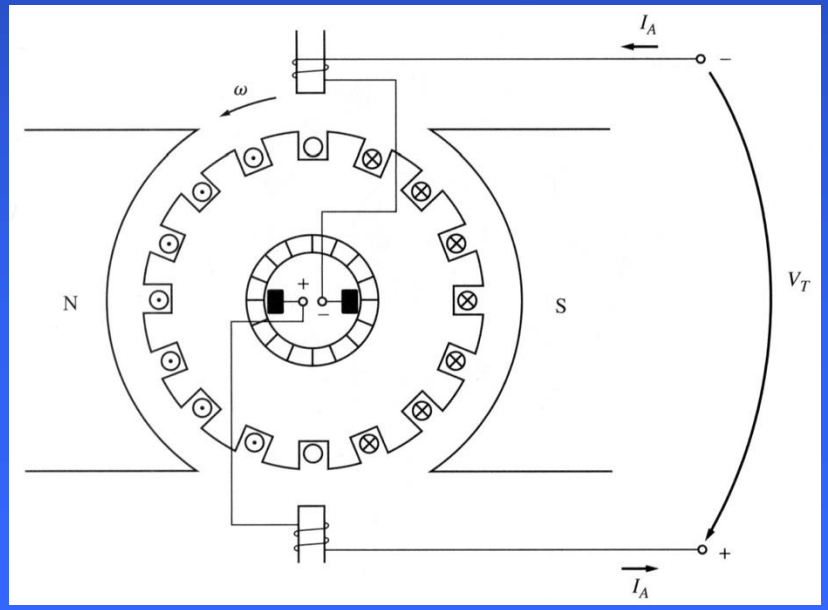


Solutions to the problems with commutation

1. Commutating poles or interpoles

To avoid sparkling at the brushes while the machine's load changes, instead of adjusting the brushes' position, it is possible to introduce small poles (commutating poles or interpoles) between the main ones to make the voltage in the commutating wires to be zero. Such poles are located directly over the conductors being commutated and provide the flux that can exactly cancel the voltage in the coil undergoing commutation. Interpoles do not change the operation of the machine since they are so small that only affect few conductors being commutated. Flux weakening is unaffected.

Interpole windings are connected in series with the rotor windings. As the load increases and the rotor current increases, the magnitude of neutral-plane shift and the size of *Ldi/dt* effects increase too increasing the voltage in the conductors undergoing commutation.



Solutions to the problems with commutation

However, the interpole flux increases too producing a larger voltage in the conductors that opposes the voltage due to neutral-plane shift. Therefore, both voltages cancel each other over a wide range of loads. This approach works for both DC motors and generators.

The interpoles must be of the same polarity as the next upcoming main pole in a generator;

The interpoles must be of the same polarity as the previous main pole in a motor.

The use of interpoles is very common because they correct the sparking problems of DC machines at a low cost. However, since interpoles do nothing with the flux distribution under the pole faces, flux-weakening problem is still present.

Solutions to the problems with commutation

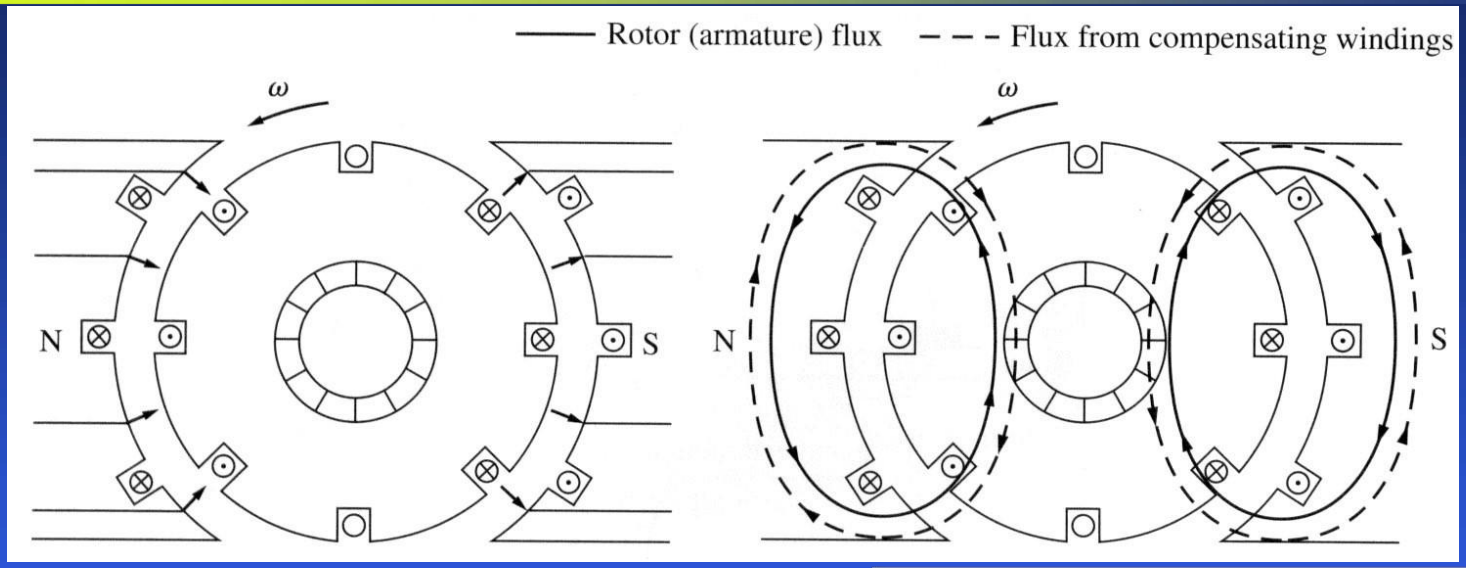
2. Compensating windings

The flux weakening problem can be very severe for large DC motors. Therefore, compensating windings can be placed in slots curved in the faces of the poles parallel to the rotor conductors.

These windings are connected in series with the rotor windings, so when the load changes in the rotor, the current in the compensating winding changes too…

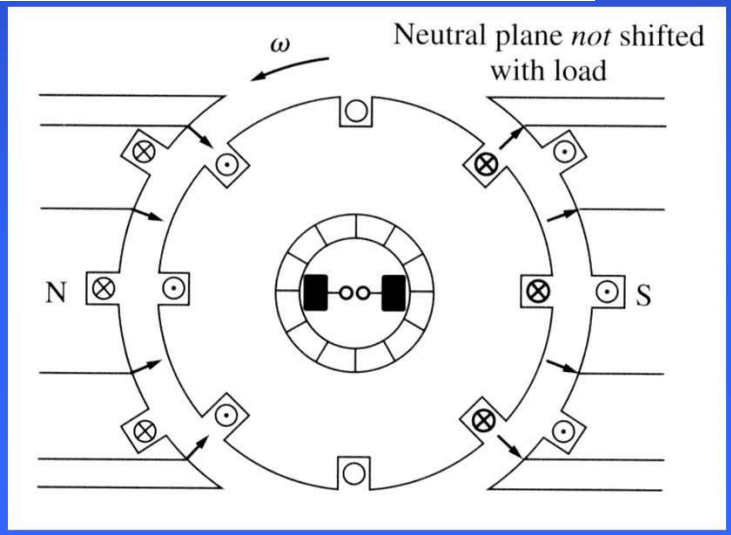
Solutions to the problems with commutation

Pole flux



Rotor and comp. fluxes

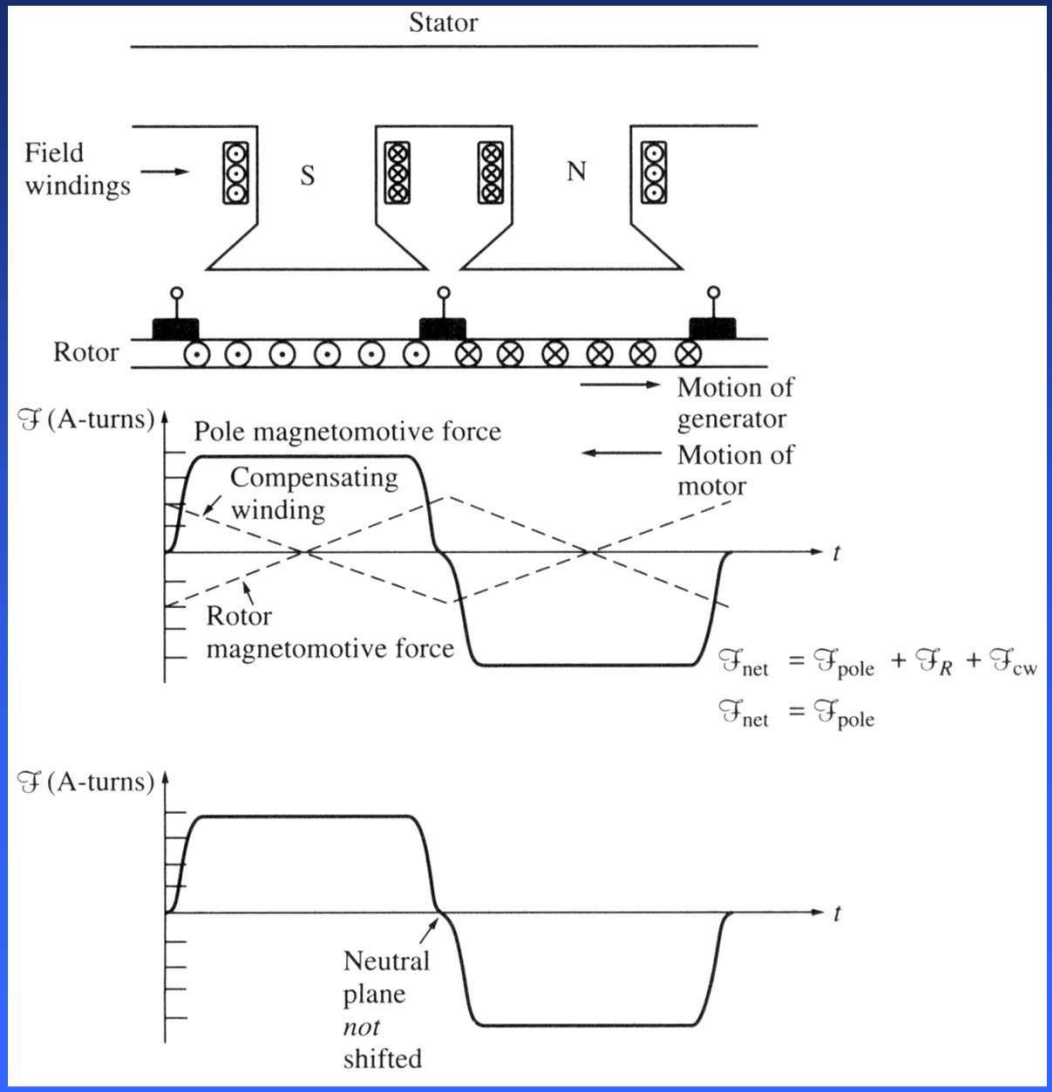
Sum of these three fluxes equals to the original pole flux.



Solutions to the problems with commutation

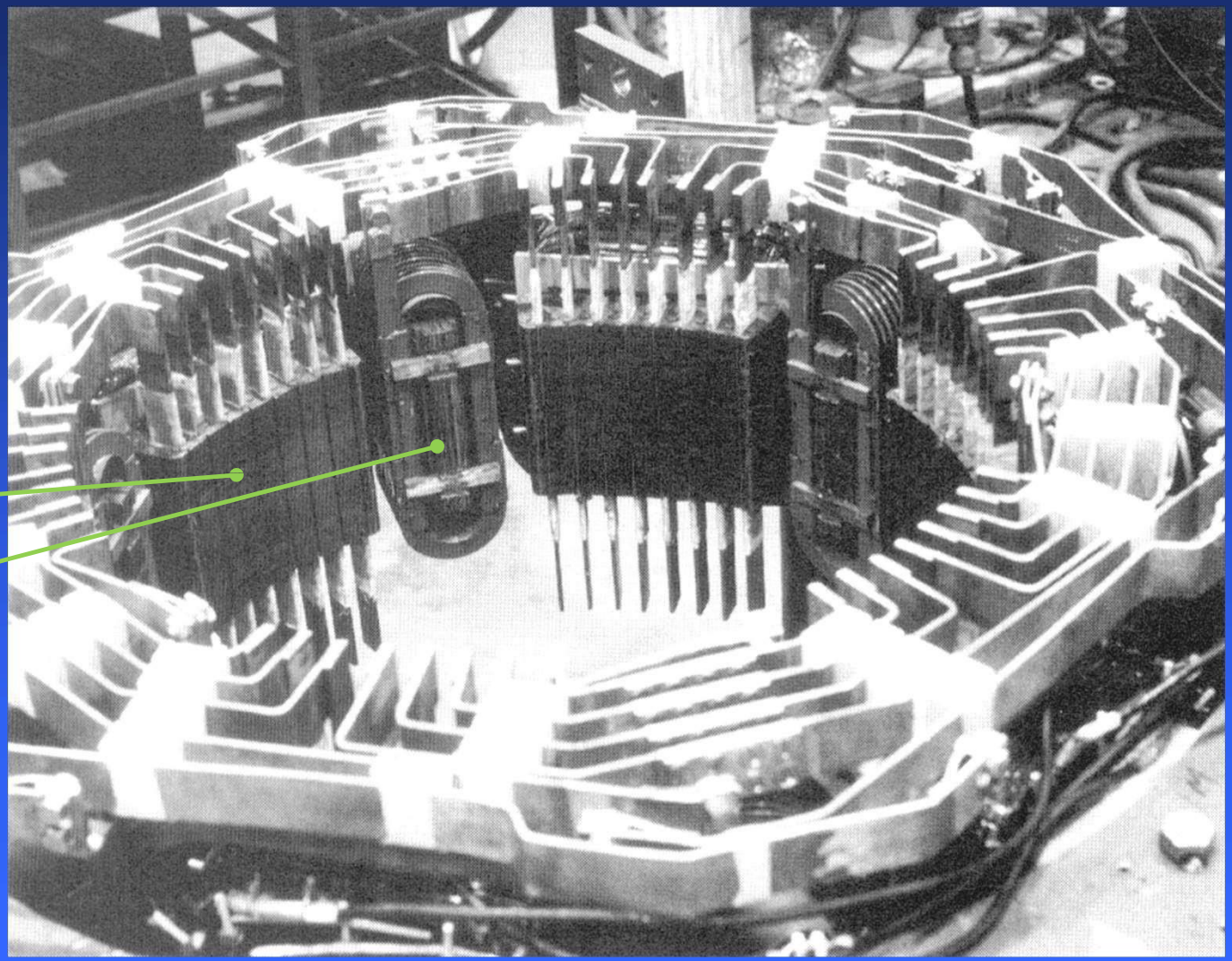
The mmf due to the compensating windings is equal and opposite to the mmf of the rotor. These two mmfs cancel each other, such that the flux in the machine is unchanged.

The main disadvantage of compensating windings is that they are expensive since they must be machined into the faces of the poles. Also, any motor with compensative windings must have interpoles to cancel *L di/dt* effects.



Solutions to the problems with commutation

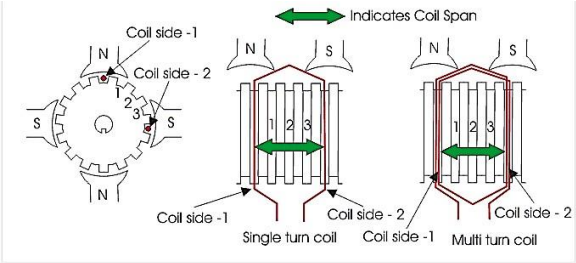
A stator of a six-pole DC machine with interpoles and compensating windings.



Commutation and Armature Construction in Real DC Machines

Rotor Coils:

- Normally, a coil spans **180 electrical degrees**.
- This means that when **one side** is **under the center** of a given magnetic pole, the **other side** is **under the center** of a pole of **opposite polarity**.



Source: https://electricalstudy.sarutech.com

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Commutation and Armature Construction in Real DC Machines

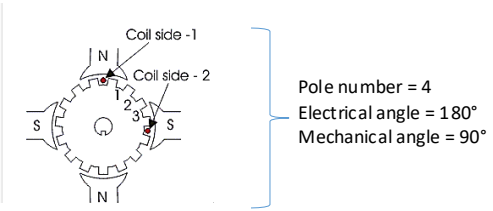
Rotor Coils:

- The relationship between the **electrical angle** and **mechanical angle** in a machine is given by

$$\theta_e = \frac{P}{2}\theta_m$$

where θ_e = electrical angle, in degrees
θ_m = mechanical angle, in degrees
P = number of magnetic poles on the machine

Example:



Pole number = 4
Electrical angle = 180°
Mechanical angle = 90°

Source: https://electricalstudy.sarutech.com

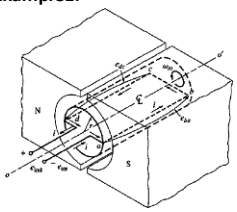
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Commutation and Armature Construction in Real DC Machines

Full-Pitch Coil:

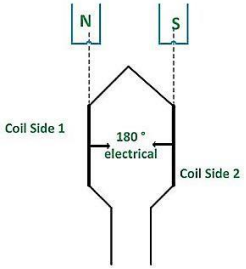
- If a coil spans **180 electrical degrees**, the voltages in the conductors on either side of the coil will be exactly the **same in magnitude** and **opposite in direction** at all times.
- Such a coil is called "***full-pitch coil***".

Example1:



Full-Pitch Coil

Example2:



Full-Pitch Coil

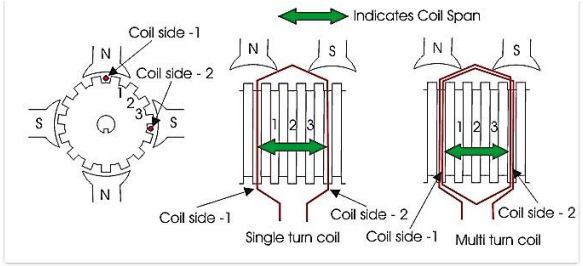
65

Commutation and Armature Construction in Real DC Machines

Full-Pitch Coil:

- If a coil spans **180 electrical degrees**, the voltages in the conductors on either side of the coil will be exactly the same in magnitude and opposite in direction at all times.
- Such a coil is called "***full-pitch coil***".

Example3:



Full-Pitch Coil

Source: https://electricalstudys.arutech.com

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Commutation and Armature Construction in Real DC Machines

Fractional-Pitch Coil:

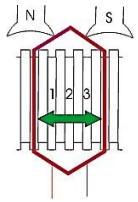
- Sometimes a coil is built that spans **less than 180 electrical degrees**.
- Such a coil is called "***fractional-pitch coil***".
- The rotor winding wound with **fractional-pitch coils** is called a "**chorded winding**"
- The **amount of chording** in a winding is described by a "**pitch factor, *p*"**, which is defined by the following equation.
- **Fractional-pitch** coil is used in DC machines to **improve commutation**.

$$p = \frac{\text{electrical angle of coil}}{180^{o}} \times 100\%$$

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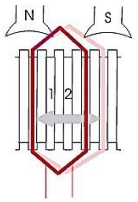
Commutation and Armature Construction in Real DC Machines

Comparison of Full-Pitch Coil and Fractional-Pitch Coil:



Full-Pitch Coil

$p = 100\%$



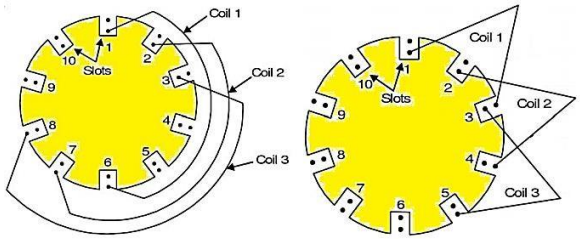
Fractional-Pitch Coil

$p = 80\%$

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Commutation and Armature Construction in Real DC Machines

Comparison of Full-Pitch Coil and Fractional-Pitch Coil:



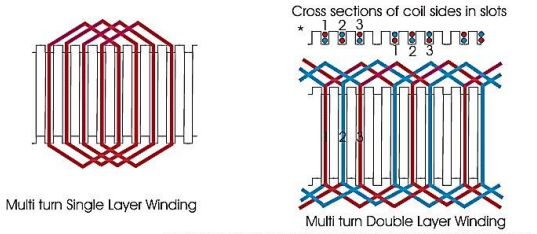
Full-Pitch Coil *Fractional-Pitch Coil*

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Commutation and Armature Construction in Real DC Machines

Multi-Winding Coil:

- Most rotor windings are **double-layer windings**.
- This means that sides from **two different coils** are inserted **into each slot**.
- One side of each coil will be at the **bottom of its slot**, and the other side will be at the **top of its slot**.
- In this fashion, all the windings are woven together, **increasing the mechanical strength and uniformity** of the final structure.



Multi turn Single Layer Winding

Multi turn Double Layer Winding

[*In Double layer winding, 1,2,3 are marked to only three coils among many]

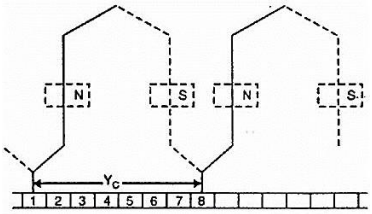
Source: https://electricalstudysarutech.com

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Commutation and Armature Construction in Real DC Machines

Connections to the Commutator Segments:

- Once the **windings** are installed in the **rotor slots**, they must be connected to the **commutator segments**.
- There are a number of ways in which these connections can be made.
- There are also different winding arrangements which result have different advantages and disadvantages.
- The **distance** (***in number of segments***) between the commutator segments to which **the two ends of a coil** are connected is called "***commutator pitch, Yc***".



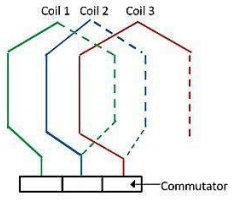
commutator pitch, Yc = 8

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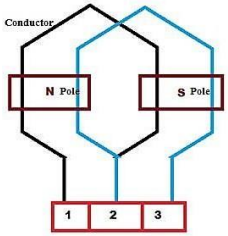
Commutation and Armature Construction in Real DC Machines

Progressive Winding:

- If the end of a coil is connected to a commutator segment **ahead** of the one its beginning is connected to, the winding is called a "**progressive winding**".



Source: https://circuitglobe.com



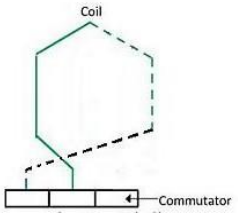
Source: http://electrialstandards.blogspot.com

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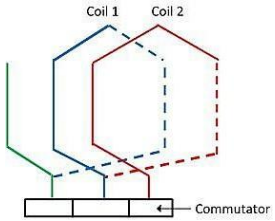
Commutation and Armature Construction in Real DC Machines

Retrogressive Winding:

- If the end of a coil is connected to a commutator segment **behind** the one its beginning is connected to, the winding is called a "**retrogressive winding**".



Source: https://circuitglobe.com



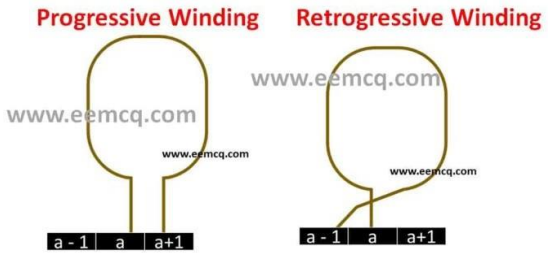
Source: https://circuitglobe.com

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Commutation and Armature Construction in Real DC Machines

Comparison of Progressive and Retrogressive Windings:

- If everything else is identical, the direction of rotation of a **progressive-wound rotor** will be **opposite** to the direction of rotation of a **retrogressive-wound rotor**.

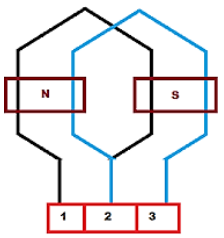


Source: www.eemcq.com

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Commutation and Armature Construction in Real DC Machines

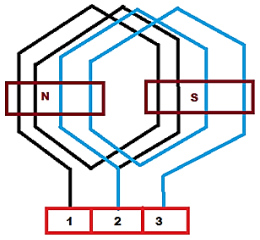
- **Rotor (*armature*) windings** are further classified according to the **plex** of their windings.
- A **simplex rotor winding** is a single, complete, closed winding wound on a rotor.
- A **duplex rotor winding** is a rotor with two complete and independent sets of rotor windings.
- A **triplex rotor winding** will have three complete and independent sets of rotor windings.
- Collectively, all armatures with **more than one set of windings** are said to have "***multiplex windings***".



Progressive Winding

Simplex lap winding

Source: https://www.electrical4u.com



Progressive Winding

Duplex lap winding

Source: https://www.electrical4u.com

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Commutation and Armature Construction in Real DC Machines

Lap Winding and Wave Winding:

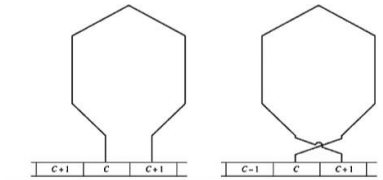
- **Armature windings** are also classified according to the **sequence of their connections** to the **commutator segments**.
- There are **two basic sequences** of armature winding connections: **"Lap Winding"** and **"Wave Winding"**.
- In addition, there is a **third type of winding**, called a **"Frog-leg winding"**, which combines **lap winding** and **wave winding** on a single rotor.
- Each one has its own **advantages** and **disadvantages**.

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Commutation and Armature Construction in Real DC Machines

Lap Winding:

- The simplest type of winding construction used in modem dc machines is the **simplex series** or **lap winding**.
- A **simplex lap winding** is a rotor winding consisting of coils containing one or more turns of wire **with the two ends of each coil coming out at** adjacent commutator segments.
- If the end of the coil is connected to the segment **after the segment** that the beginning of the coil is connected to, the winding is a **progressive lap winding**.
- If the end of the coil is connected to the segment **before the segment** that the beginning of the coil is connected to, the winding is a **retrogressive lap winding**.

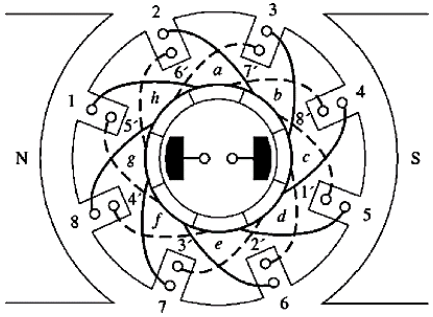


***progressive** lap winding **retrogressive** lap winding*
*commutator pitch, **Yc = 1** commutator pitch, **Yc = -1***

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Commutation and Armature Construction in Real DC Machines

Lap Winding Example:



A simple two-pole lap-wound dc machine

Lap wound rotor
Pole number = 2
Slot number = 8
Pitch factor = 100%
Commutator segment number = 8
Brush number = 2
Coil number = 8
Conductor number = 2x8 = 16

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Commutation and Armature Construction in Real DC Machines

Lap Winding:

- An interesting feature of **simplex lap windings** is that there are as **many parallel current paths** through the machine as there are **poles on the machine**.

The number of parallel current paths on the armature = the number of poles

- If **C** is the ***number of coils*** and ***commutator segments*** present in the rotor and **P** is the ***number of poles*** on the machine, then there will be **C/P *coils*** in each of the **P *parallel current paths*** through the machine.
- Also it is important to express that **P** current paths also requires that there be as **many brushes** on the machine as there are **poles** in order to tap all the current paths.

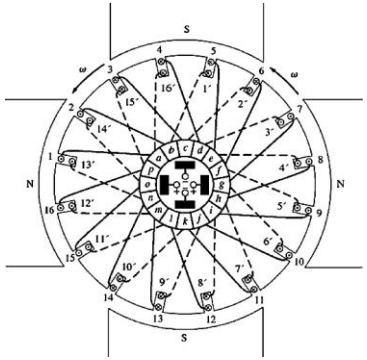
The number of brushes = the number of poles = the number of parallel current paths on the armature

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Commutation and Armature Construction in Real DC Machines

Lap Winding:

- There are **four current paths** through the rotor, each having an equal voltage for the DC machine as shown in the figure.
- The fact that there are **many current paths** in a multi pole machine makes the lap winding an ideal choice for fairly **low-voltage, high-current DC machines**.
- Since the **required high currents can be split** among the **several different current paths**.

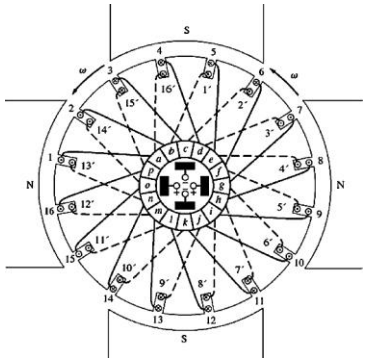


A simple four-pole lap-wound dc machine

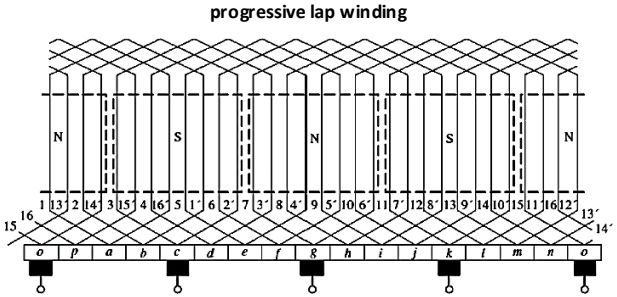
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Commutation and Armature Construction in Real DC Machines

Lap Winding:



A simple four-pole lap-wound dc machine



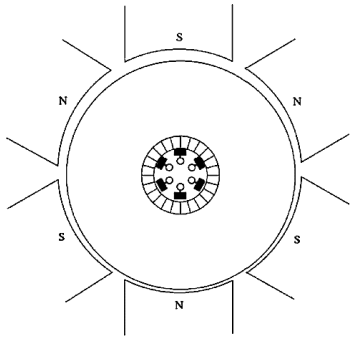
The rotor winding diagram of this machine

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Commutation and Armature Construction in Real DC Machines

Possible Problem in Lap-Wound Machines:

- Having **many parallel paths** in lap-wound machines can lead to a **serious problem**.
- As seen in the figure, **because of long usage**, there has been **slight wear on the bearings of this machine**, and the lower wires are closer to their pole faces than the upper wires are.
- As a result, there is a **larger voltage** in the current paths involving wires under the lower pole faces than in the paths involving wires under the upper pole faces.

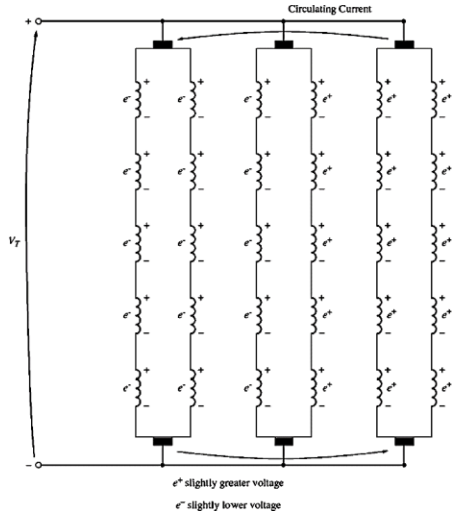


A six-pole dc motor showing the effects of bearing wear. Notice that the rotor is slightly closer to the lower poles than it is to the upper poles.

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Possible Problem in Lap-Wound Machines:

- Since **all the paths are connected in parallel**, the result will be a **circulating current** flowing out some of the brushes in the machine and back into others, as shown in the figure.
- **Circulating current** is **not good** for the machine.
- Since the winding resistance of a rotor circuit is so small, a very tiny imbalance among the voltages in the parallel paths will cause **large circulating currents through the brushes**.
- **Large circulating currents** will cause **serious heating problems** and **decrease the efficiency of the DC machine**.



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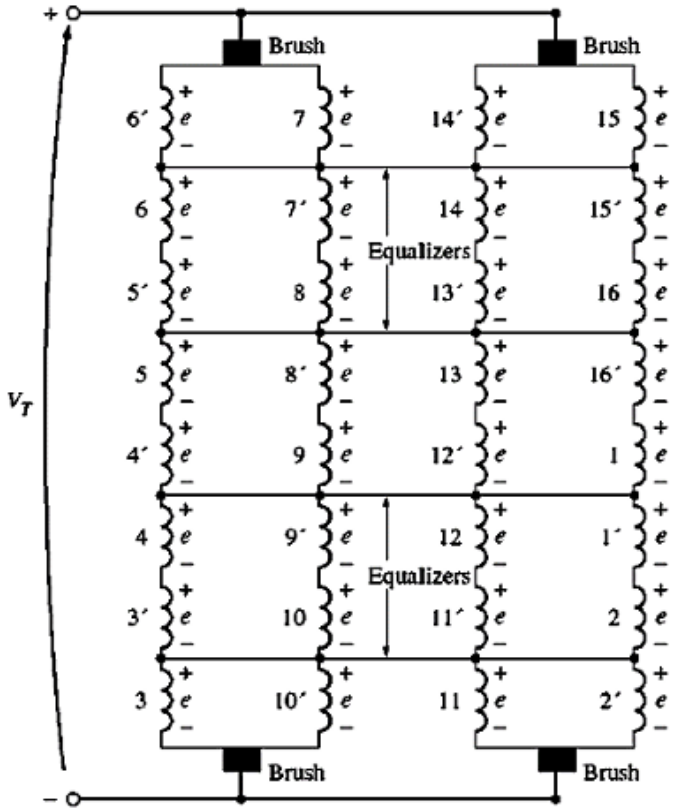
Commutation and Armature Construction in Real DC Machines

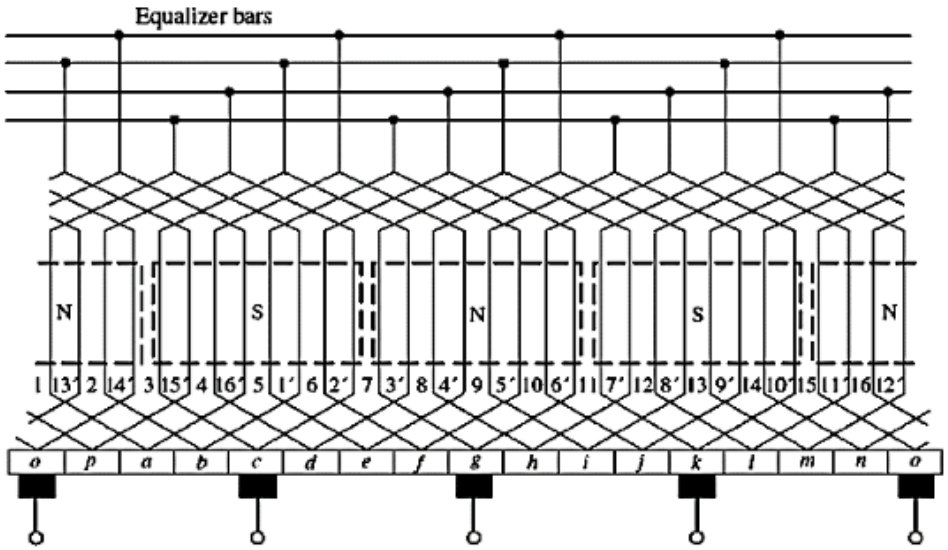
The Solution to Circulating Current Problem:

- The problem of circulating currents within the parallel paths of a machine **with four or more poles** can **never** be entirely resolved, but it **can be reduced** somewhat by ***equalizers or equalizing windings.***
- **Equalizers** are **bars** located **on the rotor** of a lap-wound dc machine that **short** together points at the **same voltage level** in the different parallel paths.
- The effect of this shorting is to cause any circulating currents that occur to now inside the small sections of windings thus shorted together and **to prevent** this circulating current from flowing through the brushes of the machine.

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Equalizer Connection for the four-pole machine





Commutation and Armature Construction in Real DC Machines

Lap Winding

- In general, for an ***m-plex*** **lap winding, the commutator pitch, yc** is

$$y_c = \pm m$$

+ for progressive winding
- for retrogressive winding

m=1 for simplex lap-winding
m=2 for dublex lap-winding
m=3 for triplex lap-winding

- **The number of parallel current paths** for a ***P-pole*** **lap winding machine** is

$$a = mP$$

m=1 for simplex lap-winding
m=2 for dublex lap-winding
m=3 for triplex lap-winding
P is the pole number

Commutation and Armature Construction in Real DC Machines

If a lap winding is duplex, then there are two completely independent windings wrapped on the rotor, and every other commutator segment is tied to one of the sets.

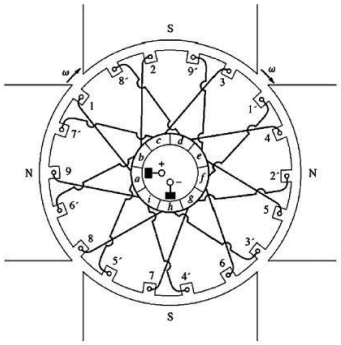
Therefore, an individual coil ends on the second commutator segment down from where it started, and $y_c = \mp 2$ (depending on whether the winding is progressive or retrogressive),

Since each set of windings has as many current paths as the machine has poles, there are *twice as many current paths* as the machine has poles in a duplex lap winding.

Commutation and Armature Construction in Real DC Machines

Wave Winding:

- The **series** or **wave winding** is an alternative way to connect the rotor coils to the commutator segments.
- For the figure, (*simplex wave winding*), every *other* rotor coil connects back to a commutator segment adjacent to the beginning of the first coil.
- Therefore, there are **two coils in series** between the adjacent commutator segments.
- Furthermore, since each pair of coils between adjacent segments has a side under each pole face, all output voltages are the **sum** of the effects of every pole, and there can be **no voltage imbalances**.



A simplex four-pole wave-wound dc machine.

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Commutation and Armature Construction in Real DC Machines

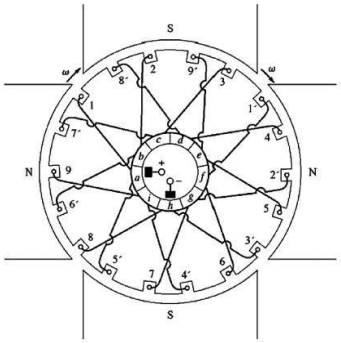
Wave Winding:

- The general expression for **commutator pitch** in any **simplex wave winding** is

$$y_c = \frac{2(C \pm 1)}{P}$$ simplex wave

where

- **C** is the number of coils on the rotor.
- **P** is the number of poles on the machine.
- The **plus sign** is associated with **progressive windings**.
- The **minus sign** is associated with **retrogressive windings.**



A simplex four-pole wave-wound dc machine.

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Commutation and Armature Construction in Real DC Machines

Wave Winding:

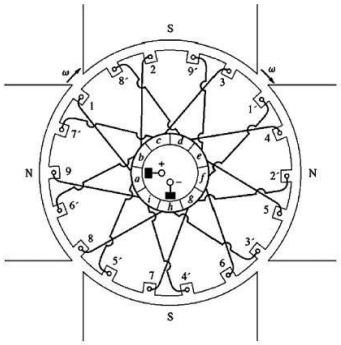
- The **number of current paths** on a **multiplex wave winding** is

$$a = 2m \qquad \text{multiplex wave}$$

where

- **m** is the **plex number**

  m=1 for **simplex** winding
  m=2 for **dublex** winding
  m=3 for **triplex** winding

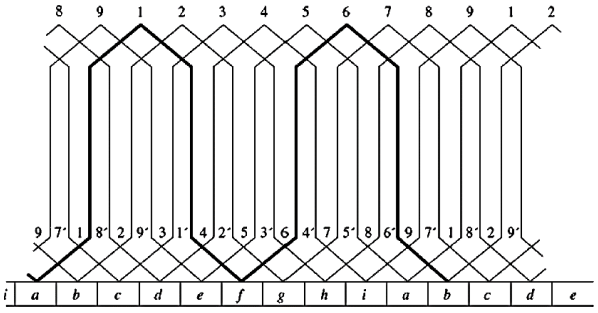


A simplex four-pole wave-wound dc machine.

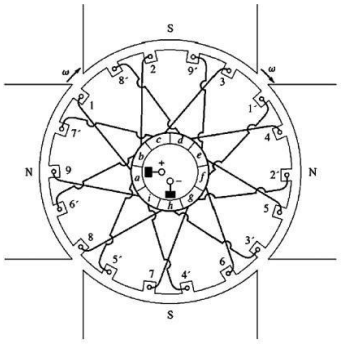
89

Commutation and Armature Construction in Real DC Machines

Wave Winding:



The rotor winding diagram

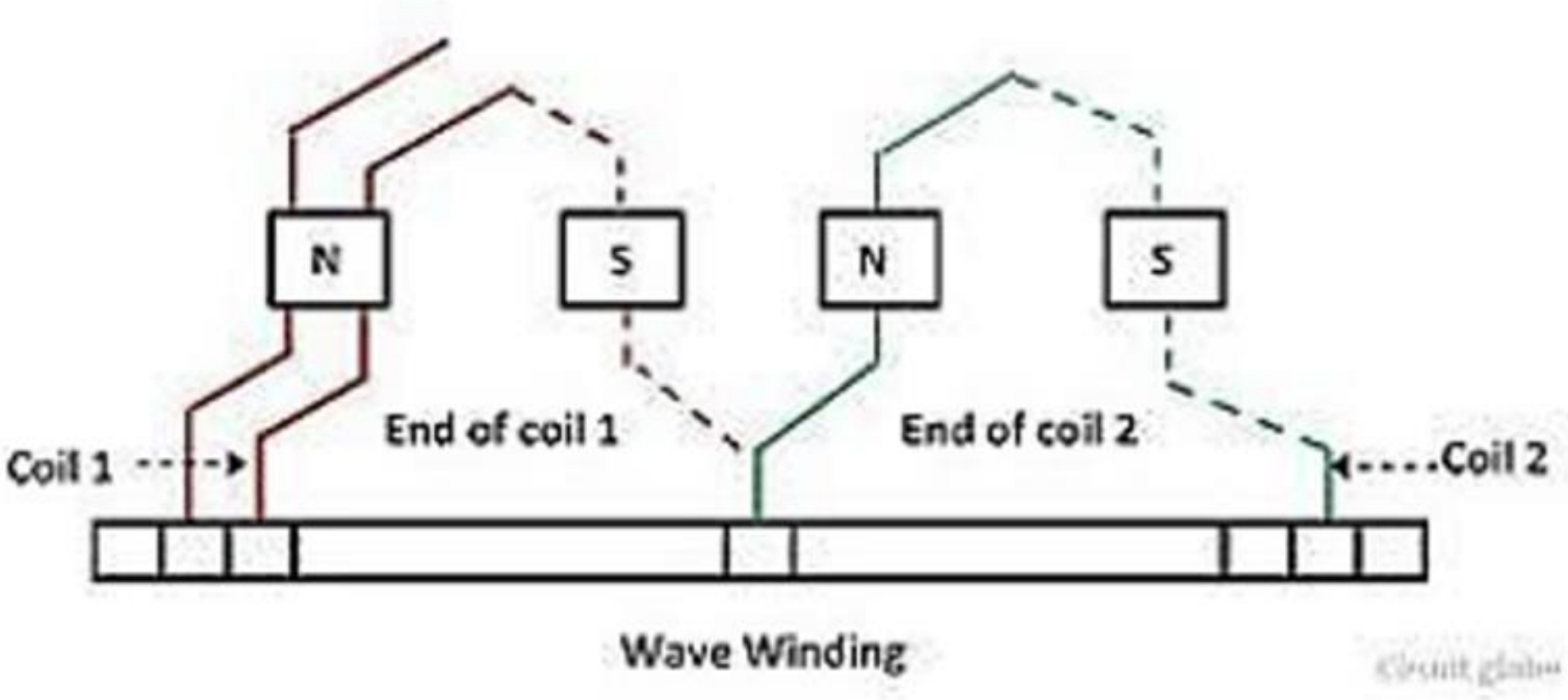


A simplex four-pole wave-wound dc machine

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Commutation and Armature Construction in Real DC Machines

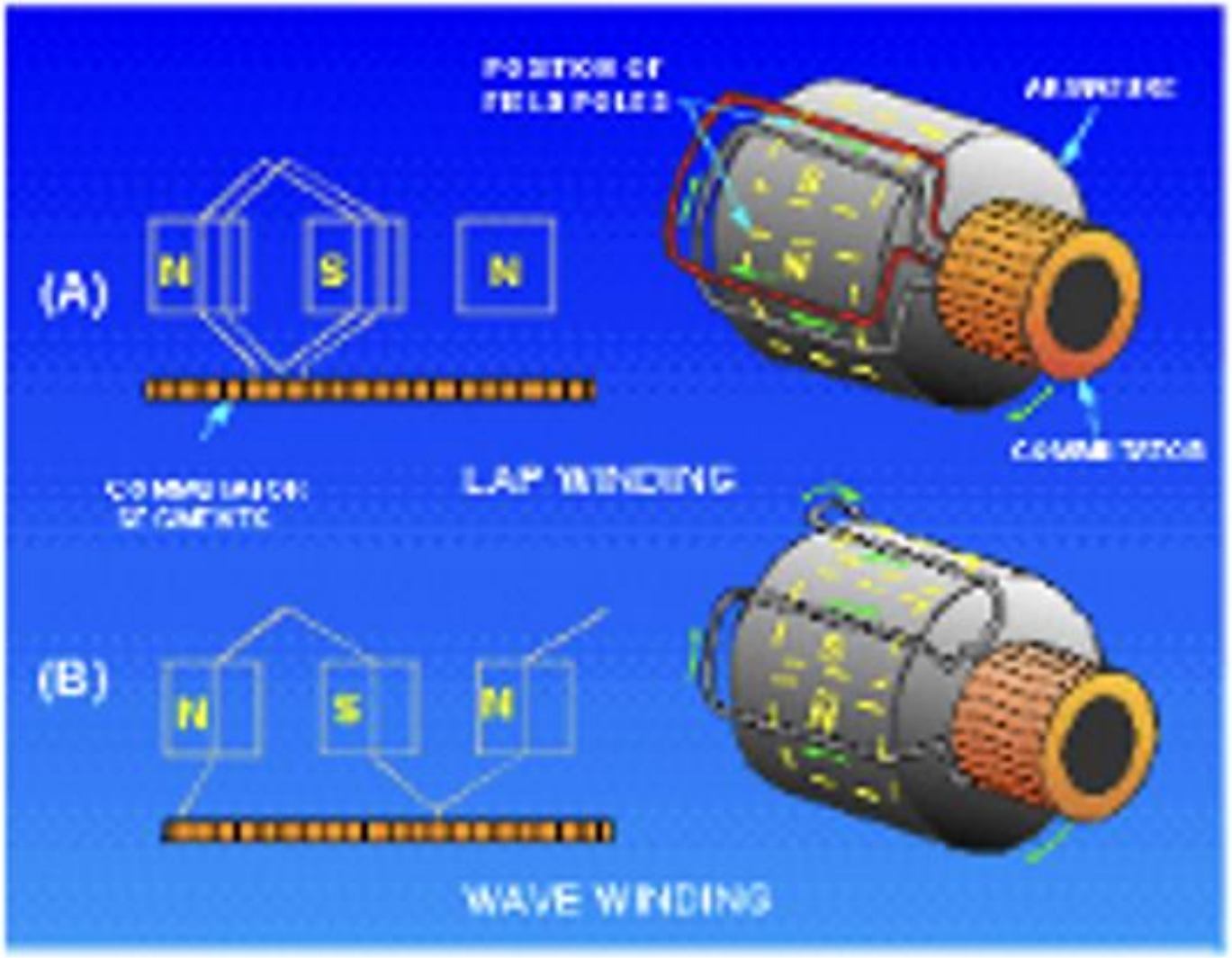
Wave Winding



Source: https://circuitglobe.com

Commutation and Armature Construction in Real DC Machines

Comparison of Lab and Wave Winding

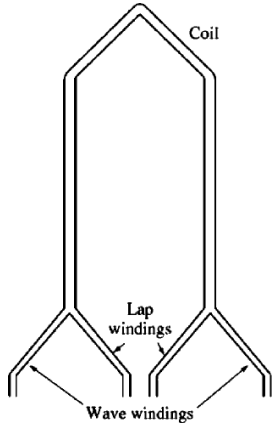


Source: https://www.ossila.com

Commutation and Armature Construction in Real DC Machines

The Frog-Leg Winding:

- The frog-leg winding or self-equalizing winding gets its name from the shape of its coils, as shown in the Figure.
- It consists of a combination of a **lap winding** and a **wave winding**.
- **Wave windings** can function as ***equalizers*** for the lap winding.
- Since the ***equalizers*** in an **ordinary lap winding** are connected at points of equal voltage on the windings, **wave windings** reach between points of **essentially equal voltage** under successive pole faces of the same polarity, which are the same locations that equalizers tie together.



A frog-leg or self-equalizing winding coil.

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Commutation and Armature Construction in Real DC Machines

The Frog-Leg Winding:

- The number of **current paths** present in a **frog-leg winding** is

$$a = 2Pm_{lap}$$

where **P** is the **number of poles** on the machine, and
$\mathbf{m_{lap}}$ is the **plex** of the lap winding.

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Power flow and losses in DC machines

Unfortunately, not all electrical power is converted to mechanical power by a motor and not all mechanical power is converted to electrical power by a generator…

The efficiency of a DC machine is:

$$\eta = \frac{P_{out}}{P_{in}} \cdot 100\% \qquad (5.36.1)$$

or

$$\eta = \frac{P_{in} - P_{loss}}{P_{in}} \cdot 100\% \qquad (5.36.2)$$

The losses in DC machines

There are **five** categories of losses occurring in DC machines.

1. Electrical or copper losses – the resistive losses in the armature and field windings of the machine.

Armature loss:

$$P_A = I_A^2 R_A \qquad (5.37.1)$$

Field loss:

$$P_F = I_F^2 R_F \qquad (5.37.2)$$

Where I_A and I_F are armature and field currents and R_A and R_F are armature and field (winding) resistances usually measured at normal operating temperature.





The losses in DC machines

2. Brush (drop) losses – the power lost across the contact potential at the brushes of the machine.

$$P_{BD} = V_{BD} I_A \qquad (5.38.1)$$

Where I_A is the armature current and V_{BD} is the brush voltage drop. The voltage drop across the set of brushes is approximately constant over a large range of armature currents and it is usually assumed to be about 2 V.

Other losses are exactly the same as in AC machines…

The losses in DC machines

3. Core losses – hysteresis losses and eddy current losses. They vary as B^2 (square of flux density) and as $n^{1.5}$ (speed of rotation of the magnetic field).

4. Mechanical losses – losses associated with mechanical effects: friction (friction of the bearings) and windage (friction between the moving parts of the machine and the air inside the casing). These losses vary as the cube of rotation speed n^3.

5. Stray (Miscellaneous) losses – losses that cannot be classified in any of the previous categories. They are usually due to inaccuracies in modeling. For many machines, stray losses are assumed as 1% of full load.

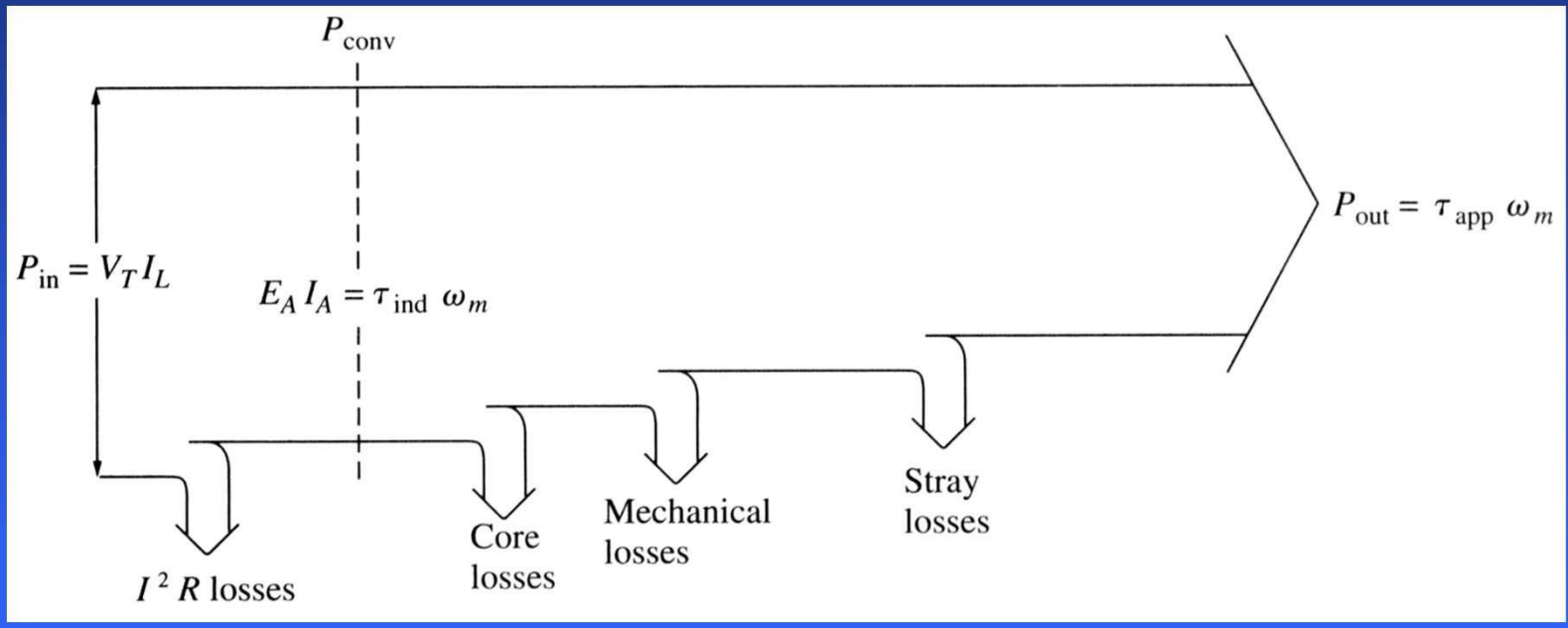




The power-flow diagram

On of the most convenient technique to account for power losses in a machine is the power-flow diagram.

For a DC motor:



Electrical power is input to the machine, and the electrical and brush losses must be subtracted. The remaining power is ideally converted from electrical to mechanical form at the point labeled as P_{conv}.

The power-flow diagram

The electrical power that is converted is

$$P_{conv} = E_A I_A \tag{5.41.1}$$

And the resulting mechanical power is

$$P_{conv} = \tau_{ind} \omega_m \tag{5.41.2}$$

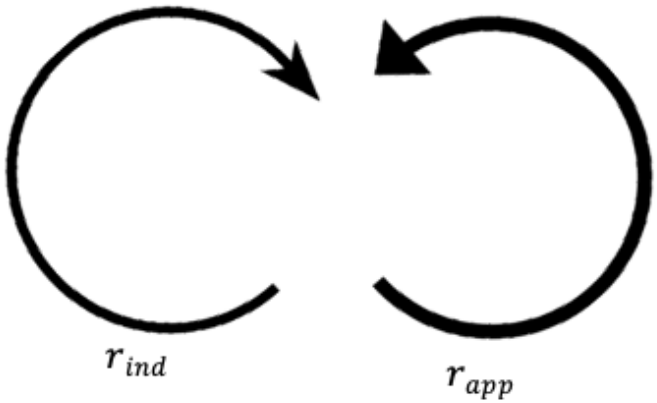
After the power is converted to mechanical form, the stray losses, mechanical losses, and core losses are subtracted, and the remaining mechanical power is output to the load.





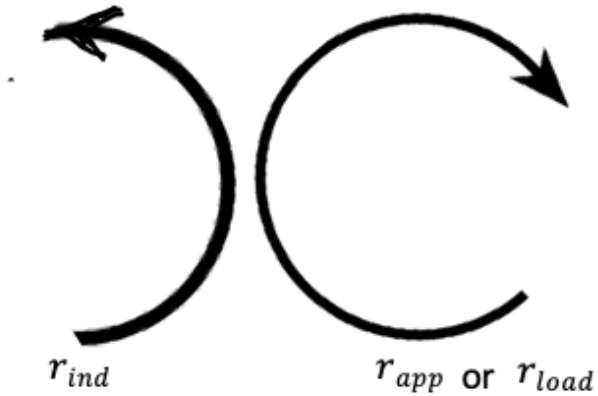
Comparison of Torques for DC Motors and Generators

For DC Generator:



$$\tau_{\text{ind}} = K\phi I_A$$

For DC Motor:



$$\tau_{\text{ind}} = K\phi I_A$$